

# The Fertility, Marriage, and Gender Equality Quandary

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## Abstract

This paper shows that the modern family transition faces a three-way constraint: societies can generally achieve two of high fertility, dual parenthood, and gender income equality, but rarely all three. To explain this fact, I develop an equilibrium marriage-market model in which marriage bundles income transfers with shared fertility. While it is challenging for conventional policies to reconcile the three objectives, a surprising result is that redistributing childcare from wives to husbands achieves the goal – but only when both spouses gain. A calibration to Mexico from 1990 to 2015 quantifies the relative contribution of gender-neutral and gender-biased technological changes.

**JEL classification:** D13, J11, J12, J13, J16

**Keywords:** Gender equality, family structure, fertility

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# 1 Introduction

Three demographic transformations have reshaped advanced economies over the past half-century. Fertility has fallen, often well below replacement. Stable two-parent families have grown less common, and a rising share of children now spend part of their childhood in single-parent households. Meanwhile, gender gaps in education and earnings have narrowed substantially, and in some settings reversed (Goldin, 2014).

Each transformation has motivated a distinct policy agenda. Concerns about pension sustainability (Bongaarts, 2004) and long-run economic growth (Jones, 2022) have prompted pro-fertility measures ranging from baby bonuses to subsidized childcare. A long-standing interest in family stability, rooted partly in cultural considerations and partly in evidence on the consequences of single parenthood for children’s outcomes (Autor et al., 2019; Wasserman, 2020), has motivated pro-marriage policies such as joint taxation. Efforts to promote gender equality, meanwhile, seek both to advance equal opportunity and to correct the misallocation of human capital, whose aggregate costs can be substantial (Hsieh et al., 2019). These agendas are usually discussed separately. Yet the decisions they seek to influence are closely connected: having children changes parents’ market work, earning opportunities shape the attraction of marriage, and marriage changes the resources available for raising children. When does progress toward one objective support the others, and when does it come at their expense? I address this question in four steps.

First, I document that the three outcomes stand in a genuinely three-way tension. Across four datasets that differ widely in geography, time span, and level of aggregation—an OECD panel spanning 1970 to 2017, a global panel of 95 countries from 1990 to 2015, U.S. state-level observations in 2023, and 656 U.S. commuting zones in 2000—societies that succeed on any two of high fertility, low single parenthood, and gender income equality almost invariably fall short on the third. The share of observations attaining all three ranges from 2.5 to 7.6 percent, well below the 12.5 percent benchmark implied by statistical independence. Mutual information analysis shows that the three outcomes exhibit synergy: the dependence between any two of them strengthens once the third is conditioned on. The three variables jointly carry information that no pair of them carries alone, so the joint structure cannot be recovered from separate pairwise analyses.

Second, I develop a tractable equilibrium marriage-market model that connects the three outcomes through income sharing and the time cost of children. Men and women draw idiosyncratic tastes for marriage. Marriage bundles an income transfer from husband to wife with shared fertility: the wife chooses fertility conditional on the transfer, and the husband evaluates the resulting consumption–fertility bundle when deciding whether to marry. The transfer then adjusts to balance participation on the two sides of the market. Fertility and the transfer are therefore jointly determined.

The model weaves together three outcomes of interest: aggregate fertility, marriage, and the

gender income gap. When mothers provide more childcare, higher fertility reduces relative female labor supply. At the same time, the income-sharing value of marriage depends on women's earning capacity relative to the support available from husbands. Their interaction generates a simple yet important policy tension. A larger marriage benefit or a lower cost of children raises fertility and marriage but widens the gender income gap. Higher female wages narrow the gap but reduce fertility and marriage.

Third, I ask whether policy can loosen this trade-off. I show that three standard levers—subsidizing marriage, closing the wage gap, and lowering the overall cost of children—each improve two of the three outcomes at the expense of the third. Combining them may appear promising, but it requires delicate balancing and coordination.

An alternative policy I consider is a *redistribution* of childcare obligations from wives to husbands, an increasingly popular policy lever used in developed economies to promote gender equality in childcare (Farré and González, 2019; Patnaik, 2019). The baseline asymmetry that such policies target is well documented: mothers shoulder far more of the time burden than fathers. Viewed through a wider lens, however, this asymmetry is itself plausibly a legacy of traditional gender norms rather than what technology or preferences would have justified (Deschamps et al., 2026). Redistribution, such as through mandated paternity leaves or changes in social norms, can therefore be understood as a corrective that pushes toward a more efficient childcare allocation.

I show that under three conditions, childcare redistribution can raise fertility, marriage, and gender income equality simultaneously. The first requires redistribution to lower the wife's effective childcare cost, which leads to a rise in marital fertility. The second requires the shift in childcare to improve women's relative labor supply even after additional fertility creates more care to be provided. The third concerns the male side of the market: despite providing more childcare, the husband must gain after fertility and transfers adjust in the equilibrium.

The bilateral nature of the conditions is the paper's central theoretical point. Mothers' time constraints and fathers' willingness to enter the marital contract jointly shape the equilibrium responses to policy. Relieving women's constraints raises marriage and fertility only when the resulting contract continues to offer men a sufficiently attractive exchange between consumption and children. Bringing both sides into the analysis explains why similar policies can produce different demographic responses in various contexts.

Fourth, I bring the framework to the data by calibrating it to Mexico from 1990 to 2015.<sup>1</sup> The period features falling fertility, rising single motherhood, a reversal of the gender education gap, and narrowing gender earnings gaps. With four estimated parameters, the model closely tracks fertility and the child-based measure of family structure. Its out-of-sample fertility prediction for

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<sup>1</sup>Mexico offers a long time series, substantial wage changes during its demographic transition, and a trajectory broadly representative of transitions in many middle-income economies.

2023 is 1.93, compared with the data value of 1.91 used in the quantitative exercise. A structural decomposition attributes roughly half of the fertility and family-structure changes to gender-neutral productivity growth and most income convergence to declining gender-biased productivity together with human-capital convergence. Applying the calibrated model to the paper’s central policy experiment, I find that the conditions for childcare redistribution to improve all three outcomes are not satisfied in the early 1990s but hold from 1998 onward. The male-side participation condition is the binding margin in the earlier years, underscoring the bilateral nature of the result.

A dynamic extension closes the paper by introducing a feedback through relative human capital. Motivated by evidence that boys are more sensitive than girls to the absence of a biological father ([Sommers, 2001](#); [Autor et al., 2019](#); [Wasserman, 2020](#)), lower marriage reduces the next generation’s male relative human capital. This change puts further downward pressure on marriage and fertility. A technological shock can therefore leave a demographic legacy beyond the generation that first experiences it. The strength of the reinforcing feedback determines whether that legacy gradually fades or generates local instability.

## **Related Literature**

The paper connects several literatures. A long tradition links fertility decisions to female labor supply through the time cost of children, beginning with [Rosenzweig and Wolpin \(1980\)](#) and continuing in quantitative models such as [Erosa et al. \(2016\)](#) and [Blundell et al. \(2016\)](#). Related work emphasizes household production, childcare marketization, and parental leave ([Bar et al., 2018](#); [Greenwood, 2019](#); [Kim and Yum, 2025](#)); [Doepke et al. \(2023\)](#) and [Greenwood et al. \(2023\)](#) provide surveys. I build on this fertility–labor-supply mechanism by placing it inside a marriage market. The marital contract then adjusts to policy, linking changes in women’s constraints to men’s willingness to marry.

A second literature, originating with [Becker \(1973\)](#), studies marriage-market clearing when men and women differ in productivity and preferences. Transferable-utility matching frameworks ([Choo and Siow, 2006](#); [Chiappori, 2017](#); [Low, 2024](#)) have been used to examine divorce law, technological change, and household production ([Stevenson and Wolfers, 2007](#); [Fernández and Wong, 2014](#); [Voena, 2015](#); [Greenwood et al., 2016](#); [Santos and Weiss, 2016](#); [Kennedy and Knowles, 2024](#)). I abstract from within-gender wage heterogeneity to isolate the between-gender margin and use an imperfectly transferable utility framework with micro-founded spousal utilities, following [Bisin and Tura \(2019\)](#). Changing the transfer changes fertility as well as consumption, making the marital bundle and the spouses’ resulting utilities endogenous to the terms of income sharing.

A third literature studies how family structure shapes children’s human capital, with boys appearing particularly vulnerable to the absence of a biological parent ([Sommers, 2001](#); [Bertrand and Pan, 2013](#); [Chetty et al., 2016](#); [Autor et al., 2019](#); [Wasserman, 2020](#); [Reeves, 2022](#); [Autor et al.,](#)

2023; Frimmel et al., 2024). Kearney (2023) discusses the implications for inequality and mobility. My dynamic extension uses this evidence to motivate a feedback from current marriage to future male relative human capital.

A fourth literature studies how technological change reshapes the family (Greenwood et al., 2005a,b, 2016; Ngai and Petrongolo, 2017; Kuhn et al., 2024; Cao et al., 2024). Olivetti and Petrongolo (2016) and Feng et al. (2023) document long-run gender-gap changes, while Hsieh et al. (2019) quantify gains from improved talent allocation. I borrow the decomposition of wage changes into gender-neutral and gender-biased components and connect them to fertility, marriage, and income inequality through a common equilibrium.

Empirically, the paper uses a descriptive benchmark in the spirit of the dartboard logic of Ellison and Glaeser (1997): joint attainment is compared with the frequency implied by independent outcomes having the same observed marginal attainment rates. Rank-based comparisons and alternative measurement exercises assess how broadly the pattern extends. These tools organize the empirical motivation for studying the outcomes together; the model supplies the economic structure needed to examine their responses to policy.

The rest of the paper is organized as follows. Section 2 presents the empirical patterns. Section 3 develops the baseline model and individual-policy comparative statics. Section 4 studies childcare redistribution. Section 5 presents the calibration and decomposition. Section 6 introduces dynamic propagation, and Section 7 concludes.

## 2 Motivating Facts

This section establishes the three-way tension that motivates the theory. To do so, I combine two complementary statistical tools. Venn diagrams reveal how often all three outcomes are achieved together, and mutual information analysis tests whether the three outcomes genuinely need to be studied jointly rather than as three separate pairs.

### 2.1 Data and Methods

#### Data sources and samples

Four datasets, summarized in Table 1, provide complementary vantage points on the quandary.

The *OECD sample* offers the longest time dimension, covering 37 countries from 1970 to 2017. Its family-structure indicator is the share of births outside marriage.<sup>2</sup> This measure requires careful interpretation because a child born to unmarried parents may still live with both parents. The *World sample* addresses that limitation directly by measuring children’s co-residence with both parents,

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<sup>2</sup>See Figure A.1 for the time series plot of fertility, out-of-wedlock birth, and gender earnings gap in the sample.

| Sample      | $N$   | Period    | Family structure           | Gender equality     |
|-------------|-------|-----------|----------------------------|---------------------|
| OECD        | 721   | 1970–2017 | Out-of-wedlock births      | Median earnings gap |
| World       | 1,130 | 1990–2015 | Children with both parents | Female income share |
| U.S. States | 51    | 2023      | Single-mother households   | Median earnings gap |
| U.S. CZ     | 656   | 2000      | Single-mother households   | Earnings ratio      |

Table 1. Summary of Data Samples

*Notes:* CZ = Commuting Zones. OECD and World samples use the total fertility rate for fertility; U.S. state sample uses birth per 1000 woman for fertility; U.S. CZ sample uses the crude birth rate for fertility. OECD data are drawn from the OECD Family Database. The World sample combines UN fertility data, children’s living arrangements from [Brenøe and Wasserman \(2025\)](#), and the female labor income share from the World Inequality Database ([Chancel et al., 2022](#)). U.S. state data come from CDC Natality files and the American Community Survey. U.S. CZ data are from [Chetty et al. \(2014\)](#).

using the harmonized living-arrangement data of [Brenøe and Wasserman \(2025\)](#). It combines those observations with fertility and female labor income shares. The female labor income share is a more comprehensive measure of gender inequality than the earnings gap among workers, since it incorporates differences in participation.

For the country panels, missing values are interpolated linearly within country, between observed years. Each analysis uses observations with all three required outcomes. Every retained country-year receives equal weight, so countries with longer coverage contribute more observations to the pooled distribution. Appendix [A.5](#) examines the sensitivity to interpolation and to individual countries.

The two *U.S. samples* serve as a within-country check, holding the institutional environment roughly fixed. Cross-state variation in 2023 tests whether the trade-off survives when culture, law, and macroeconomic conditions are broadly shared, and the 656 commuting zones for the year 2000 provide a finer geographic cut. The corresponding results are reported in Appendix [A.3](#).

## Variable construction

For each sample, I split all three outcomes at their sample-specific medians. “High fertility” (F) exceeds the median total fertility rate, “dual parenthood” (M) exceeds the median share of children with both parents, or equivalently falls below the median share born out of wedlock, and “gender equality” (G) falls below the median earnings gap, or equivalently exceeds the median female labor income share. Using sample medians avoids arbitrary thresholds and yields balanced categories.

I orient each variable so that higher values correspond to the designated objective before splitting. The specific cutoffs, which vary across samples because the underlying variables differ, appear in Table [2](#). All classifications use strict inequalities: an observation exactly at its cutoff remains outside the corresponding attainment group.

The benchmark asks how often all three indicators would equal one if their observed marginal

| Sample      | High fertility (F)  | Dual parenthood (M)  | Gender equality (G)  |
|-------------|---------------------|----------------------|----------------------|
| OECD        | TFR > 1.69          | OOW < 31.4%          | Gap < 17.2%          |
| World       | TFR > 2.81          | Both parents > 75.9% | Female share > 30.2% |
| U.S. States | Births/1,000 > 56.9 | Single mom < 17%     | Gap < 25.6%          |
| U.S. CZ     | CBR > 12.99         | Single mom < 19.7%   | Ratio < 1.80         |

Table 2. Median Cutoffs for Binary Indicators

*Notes:* Displayed cutoffs are rounded; calculations use the numerical thresholds returned by the statistical routine. TFR denotes the total fertility rate, children per woman. OOW denotes the out-of-wedlock birth share. CBR denotes the crude birth rate, expressed here per 1,000 population. “Gap” is (male median earnings – female median earnings) divided by male median earnings. “Ratio” is male median earnings divided by female median earnings, so lower values correspond to the gender-equality objective.

attainment rates were combined independently. Writing  $p_F$ ,  $p_M$ , and  $p_G$  for those rates, expected joint attainment is

$$p_{111}^{\text{ind}} = p_F p_M p_G.$$

I compare this quantity with the observed joint share,  $p_{111}$ . Independent indicators with marginal attainment probabilities of one-half yield 12.5 percent. Using the observed marginals accommodates finite samples and ties, which make the exact benchmarks slightly different in the OECD and state samples. For inference, I use bootstrap replications and re-estimate all three median cutoffs in every replication.

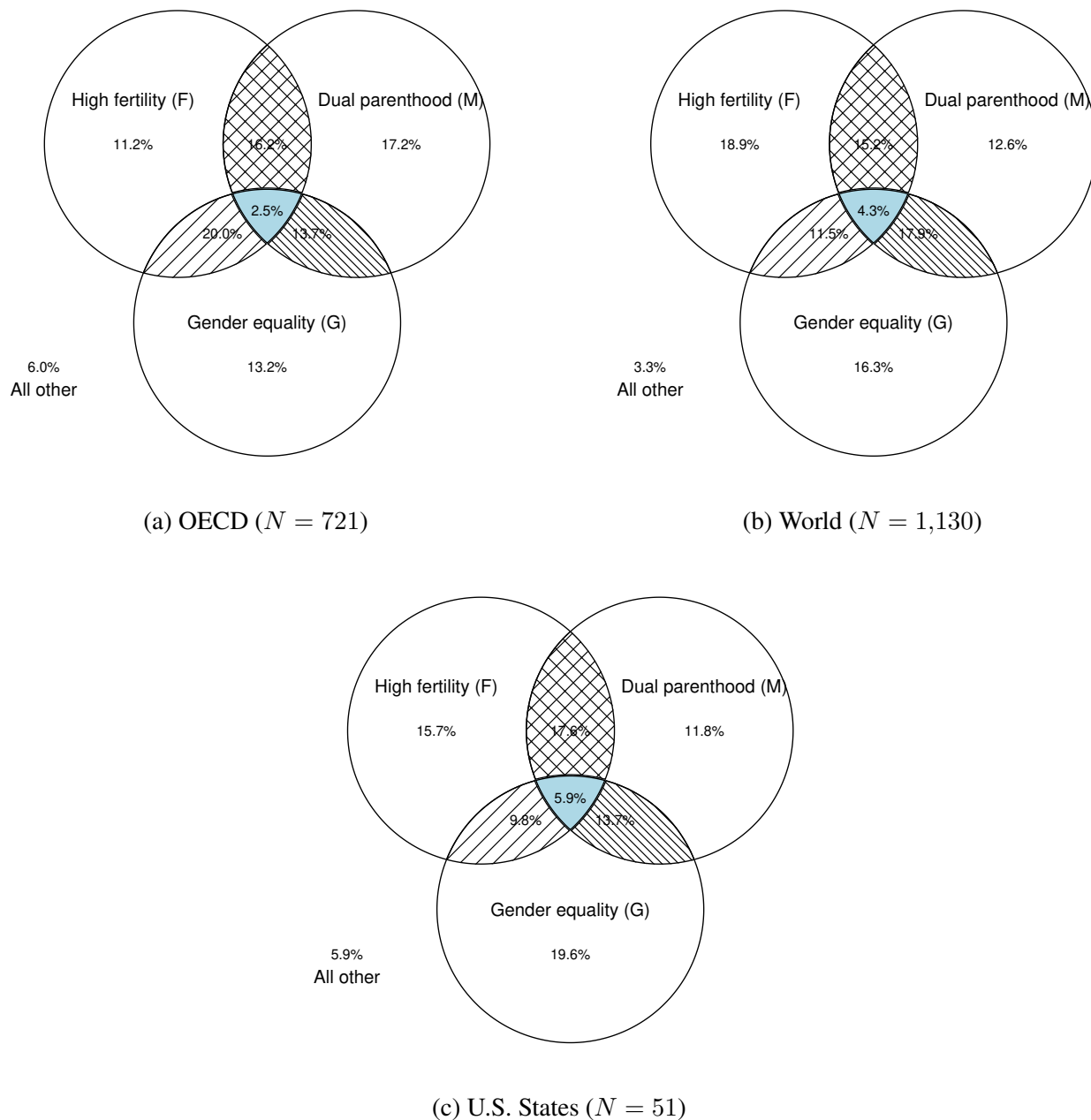
## 2.2 Results

### Achieving all three is rare

Figure 1 presents Venn diagrams for the three main samples. A remarkably uniform pattern emerges: the share of observations attaining the trinity F+M+G lies far below the 12.5 percent independence benchmark in every sample.

In the OECD panel, only 2.5 percent of country-year observations achieve the trinity—one-fifth of the independence benchmark. Country-clustered bootstrap inference yields a 95 percent percentile interval of  $[-11.79, -5.32]$  percentage points for the shortfall, and the two-sided normal-approximation  $p$ -value is below 0.001. The World sample, which uses the preferred measure of children’s co-residence with both biological parents, shows a share of 4.3 percent, with a 95 percent interval of  $[-11.46, -3.53]$  percentage points and a normal-approximation  $p$ -value below 0.001. Even the U.S. cross-state analysis, which holds institutions roughly constant, still shows only 5.9 percent. Across U.S. commuting zones, the share is 7.6 percent, with a 95 percent percentile interval of  $[-6.78, -3.16]$  (see Appendix A.3).

Figure 2 traces OECD countries’ movements across the eight categories over time. The dominant pattern is a departure from F+M toward M+G or G alone: countries typically secure gender equality



**Figure 1. Three-Way Trade-Off Across Samples**

*Notes:* Venn diagrams showing the joint distribution of high fertility (F), dual parenthood (M), and gender equality (G), each defined at sample-specific medians. Under statistical independence, the central overlap (F+M+G) would contain 12.5 percent of observations. The empirical independence benchmarks for the central overlap are 12.24 percent for OECD, 12.50 percent for World, and 11.78 percent for U.S. states. Results for U.S. commuting zones are reported in Appendix A.3.

by giving up fertility, and often give up dual parenthood in the process.

Figure 3 maps joint attainment across U.S. states. The three states that achieve the trinity—Alaska, Hawaii, and Minnesota—are geographically and demographically diverse.

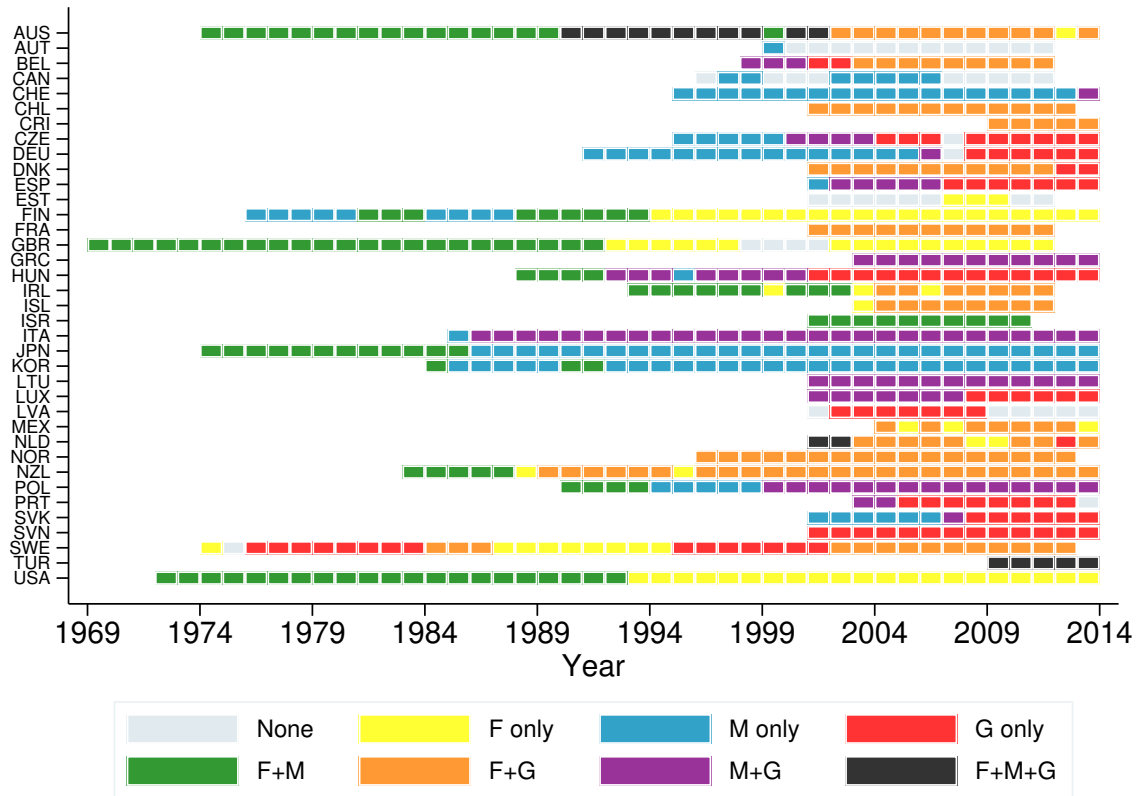


Figure 2. Transition Paths by Country (OECD Sample)

*Notes:* Temporal transitions in the three categories for each country (F = high fertility, M = dual parenthood, G = gender earnings equality). Countries are labeled by three-letter ISO codes. Data cover 37 OECD countries from 1970 to 2017. See Appendix Figure A.2 for the corresponding figure for the World sample.

### The trade-off is synergistic, not redundant

A brief primer on mutual information is helpful. Mutual information measures how much knowing one variable reduces uncertainty about another. It is measured in bits: a value of zero means statistical independence, and larger values indicate stronger dependence. Unlike a correlation coefficient, mutual information captures dependencies of any form and makes no distributional assumptions.

When three variables are involved, a natural question is whether their joint behavior can be reduced to a collection of pairwise relationships. Interaction information answers this question. It compares the dependence between two variables before and after conditioning on a third. A positive value indicates redundancy: conditioning on one variable weakens the apparent link between the other two. A negative value indicates synergy: conditioning on the third variable strengthens the relationship between the other two, so the joint distribution cannot be reconstructed from pairwise pieces alone.

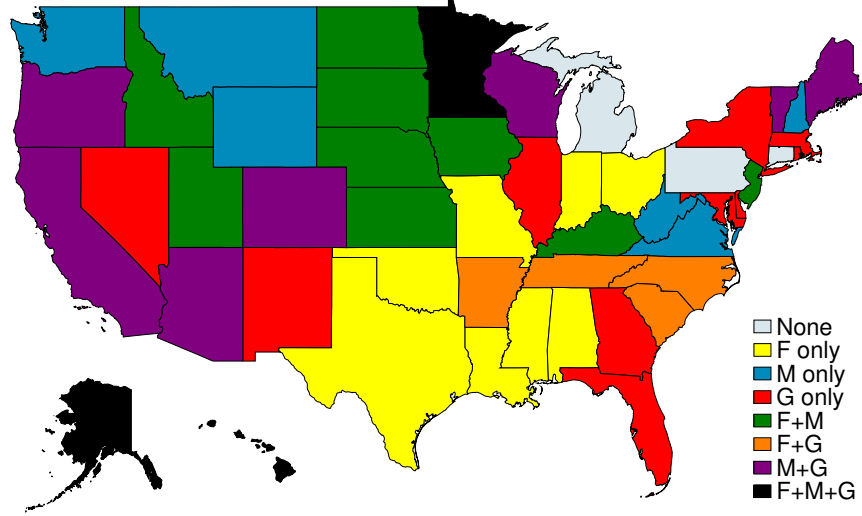


Figure 3. Joint Attainment by U.S. State (2023)

*Notes:* The map indicates which outcomes (F = high fertility, M = dual parenthood, G = gender earnings equality) each state achieves based on median thresholds. The numerical sample contains the fifty states and the District of Columbia. See Appendix Figure A.4 for the corresponding map at the commuting-zone level.

Table 3 reports these quantities for each sample. In the OECD sample, the estimate is  $-0.035$ . The World sample gives  $-0.026$  bits. The U.S. state sample gives  $-0.007$  bits. These sample estimates suggest that conditioning strengthens measured dependence.

The dependency structure can also be illustrated through conditional mutual information. In the OECD sample, the unconditional mutual information between fertility and dual parenthood is  $I(F; M) = 0.043$  bits. Conditioning on gender equality raises it to  $I(F; M | G) = 0.078$  bits. In the World sample, conditioning on dual parenthood raises the fertility–gender-equality dependence from  $I(F; G) = 0.099$  to  $I(F; G | M) = 0.125$  bits. These patterns are consistent with synergy.

| Sample      | $I(F; M)$ | $I(F; G)$ | $I(M; G)$ | Total Corr. | Interaction |
|-------------|-----------|-----------|-----------|-------------|-------------|
| OECD        | 0.043     | 0.006     | 0.081     | 0.165       | $-0.035$    |
| World       | 0.034     | 0.099     | 0.009     | 0.169       | $-0.026$    |
| U.S. States | 0.000     | 0.082     | 0.023     | 0.112       | $-0.007$    |

Table 3. Mutual Information Analysis Across Samples

*Notes:* All values are in bits. F denotes high fertility, M denotes dual parenthood, and G denotes gender income equality. Pairwise MI denotes pairwise mutual information. Total correlation summarizes the total dependence among the three variables. Interaction information measures the higher-order component: negative values indicate synergy, meaning that conditioning on any one of the three variables increases measured dependence between the other two.

## Robustness

In Appendix A.4, I split each sample at the median of GDP per capita and at the median of average years of schooling, redefining the three indicators at subsample medians to probe whether the trade-off reflects a single income or education regime. The quandary result is robust.

Appendix A.5 reports several additional checks. First, the trinity shortfall is not an artifact of using median splits. Repeating the exercise with percentile thresholds from the 25th to the 75th percentile leaves the observed-to-independence ratio below one throughout the OECD panel, the World panel, and U.S. commuting zones; the U.S. state result is also below one except in the sparsest upper-tail cutoff. I also report a continuous rank-based analogue that avoids binary thresholds altogether. For each observation, I rank fertility, dual parenthood, and gender equality after coding each variable in the direction of the corresponding policy objective, and then average the product of the three percentile ranks. This statistic is below its independence benchmark of 0.125 in every sample.

Second, the result is robust to absolute, policy-interpretable thresholds. For example, in the World sample, observations with fertility above 2.1, dual parenthood above 75 percent, and female income shares above 35 percent occur at only one-quarter of the rate implied by the observed marginals; using stricter thresholds reduces the observed-to-expected ratio below 0.10. Third, the result survives alternative measures of family structure and gender equality, including illegitimate births, single-mother shares, and alternative female income-share measures.

Finally, because the OECD and World samples are panels, I bootstrap the trinity shortfall clustering at the country level. The shortfall remains highly significant in both panels. Appendix A.6 also reports continuous interaction regressions. These regressions answer a different question from the main joint-distribution exercise: they ask whether within-country deviations around country and year means exhibit the same three-way pattern. The evidence is therefore treated as supplementary. It is strongest in U.S. commuting zones, where all three continuous interactions are negative and statistically significant, and more mixed in the country panels once country and year fixed effects absorb most of the level variation.

## 2.3 Summary

Two findings emerge from the empirical analysis. First, simultaneously achieving high fertility, widespread dual parenthood, and gender income equality is rare: the joint share ranges from 2.5 to 7.6 percent of observations, consistently below the 12.5 percent independence benchmark, and the country-panel shortfalls are statistically significant. Second, the three outcomes exhibit negative interaction information in every main sample. In the next section, I provide a theoretical framework to understand these patterns.

### 3 Model

This section develops an equilibrium marriage-market model in which fertility, marriage, and the gender income gap are determined together. The model is deliberately simple. Its purpose is to make clear how the time cost of children and the income-sharing role of marriage connect the three outcomes, and why policies directed at one can move the others in an unintended direction.

#### 3.1 Setup

Individuals are indexed by gender  $g \in \{\sigma, \varphi\}$ , with each gender comprising half the population. Utility depends on consumption  $c$  and fertility  $n$  through a constant-elasticity-of-substitution aggregator:

$$u(c, n) = \left[ (1 - \beta)c^{\frac{\rho-1}{\rho}} + \beta n^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}}, \quad \beta \in (0, 1), \quad \rho > 1. \quad (1)$$

Following [Jones and Schoonbroodt \(2010\)](#) and [Carlos Córdoba and Ripoll \(2019\)](#), I assume that consumption and fertility are substitutes. This assumption allows rising opportunity costs to reduce fertility as wage grows. Utility in (1) is also well-defined for childless individuals, with  $u(c, 0) = (1 - \beta)^{\rho/(\rho-1)}c$ .

To focus on between-gender differences, wages are uniform within each gender and strictly positive.<sup>3</sup> Let  $w^\sigma$  and  $w^\varphi$  denote male and female wages, and define the gender wage gap as  $\Gamma^w = w^\sigma / w^\varphi$ . A higher ratio indicates a larger male wage advantage. Wages are exogenous in the static model.<sup>4</sup>

#### Single individuals

Single men supply one unit of labor, consume their earnings, and remain childless:

$$V^{\sigma,s} = u(w^\sigma, 0).$$

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<sup>3</sup>Abstracting from within-gender wage and education heterogeneity shuts down assortative matching, which would require additional structure ([Chiappori, 2017](#)). Figure A.8 reports a spousal earnings gap among newly-wed, childless U.S. couples of 20 to 30 percent, compared with approximately 10 percent among young workers, motivating attention to within-couple gender differences.

<sup>4</sup>In life-cycle models such as [Erosa et al. \(2016\)](#) and [Blundell et al. \(2016\)](#), labor supply also affects subsequent human capital and wages. The static model isolates the contemporaneous labor-supply and marriage-market channels.

Single women can have children without support from the father.<sup>5</sup> They choose consumption, market work, and fertility to solve

$$V^{\varnothing,s} = \max_{c^{\varnothing,s}, l^s, n^s} u(c^{\varnothing,s}, n^s)$$

subject to

$$c^{\varnothing,s} = w^{\varnothing} l^s, \quad l^s = 1 - \chi n^s, \quad c^{\varnothing,s} \geq 0, \quad l^s \geq 0, \quad n^s \geq 0.$$

Each child requires  $\chi > 0$  units of the mother's time. A single mother therefore faces a direct choice between market consumption and children: more children leave less time to earn the income that finances consumption. Following [De La Croix and Doepke \(2003\)](#), fertility is continuous.

The setup captures the competition between consumption and fertility for household resources. Purchased childcare ([Bar et al., 2018](#)) and explicit investments in child quality ([Becker and Lewis, 1973](#)) are left for extensions. Section 4.5 discusses how outsourced care would differ from the time-cost changes studied here.

## Married individuals

Marriage changes the resources available for raising children. In the baseline, husbands provide no childcare, supply one unit of labor, and transfer a share  $\alpha$  of their earnings to their wives. The transfer captures the income-sharing function of marriage. Individuals take it as given, while its equilibrium value equates the number of men and women willing to marry.

The necessary resource restrictions are

$$-\frac{w^{\varnothing}}{w^{\sigma}} < \alpha < 1.$$

The lower bound ensures positive resources for the wife before childcare, and the upper bound leaves positive consumption for the husband. Negative transfers are permitted in principle, although the interior equilibrium characterized below has  $\alpha > 0$ .

Fertility decision rights are assigned to the wife. She chooses fertility conditional on the committed transfer, and the husband evaluates the resulting bundle when deciding whether to

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<sup>5</sup>The model abstracts from child support. Introducing enforceable support obligations would change both the resources available to single mothers and men's outside option, and would therefore require a corresponding change in the marriage-market analysis. According to the analysis by the Annie E. Casey Foundation using 2020-22 Current Population Survey data, only 23 percent of U.S. female-headed families with children under 18 receive any child support in the prior year.

marry.<sup>6</sup> His marital value is

$$V^{\sigma^g, m} = \lambda u((1 - \alpha)w^{\sigma^g}, n^m), \quad \lambda > 0.$$

The wife solves

$$V^{\varphi, m} = \max_{c^{\varphi, m}, l^m, n^m} \lambda u(c^{\varphi, m}, n^m)$$

subject to

$$c^{\varphi, m} = \alpha w^{\sigma^g} + w^{\varphi} l^m, \quad l^m = 1 - \chi n^m, \quad c^{\varphi, m} \geq 0, \quad l^m \geq 0, \quad n^m \geq 0.$$

Parameter  $\lambda$  is a utility shifter that scales the utility from marriage, reflecting factors such as social norms. In the baseline model, I assume that  $\lambda$  is gender-neutral.<sup>7</sup>

The within-marriage transfer creates an income effect for the wife: at a given wage, more support allows her to choose both more consumption and more children. A higher female wage raises income while also making time spent caring for children more expensive. With  $\rho > 1$ , the opportunity-cost effect dominates and desired fertility falls at a fixed nonnegative transfer. This is the substitution mechanism emphasized by [Greenwood et al. \(2005a\)](#). A higher male wage raises the resources available for child-rearing at a fixed positive transfer, consistent with the income channel in [Jakobsen et al. \(2022\)](#). The equilibrium analysis adds a further step by allowing the transfer itself to respond to these wage changes.

## Marriage market

At the start of the period, each individual draws independent, multiplicative taste shocks for marriage and singlehood. The shocks have a common Fréchet distribution with shape parameter  $\theta > 0$ . Individuals choose the alternative with the higher taste-adjusted value, yielding

$$\mathcal{M}^g = \frac{(V^{g, m})^\theta}{(V^{g, m})^\theta + (V^{g, s})^\theta}, \quad g \in \{\sigma^g, \varphi\}. \quad (2)$$

A higher marriage-to-singlehood utility ratio attracts more participants; a larger  $\theta$  makes participation more responsive to that ratio. Appendix C.1 derives (2) directly from the taste-shock distribution.

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<sup>6</sup>The timing commits the transfer before fertility is chosen and assigns the fertility decision to the wife. Introducing subsequent bargaining, consent, or renegotiation after children are born would require a different contracting environment.

<sup>7</sup>Allowing for gender-specific  $\lambda$  would not change the analysis as the main proposition come from comparative statics. With additional data, such as direct measures of within-marriage transfers, it would be feasible to calibrate gender-specific  $\lambda$ .

With equal masses of men and women, market clearing requires  $\mathcal{M}^{\sigma} = \mathcal{M}^{\varphi} = \mathcal{M}$ . Because the two genders have the same taste-shock distribution, this is equivalent to

$$\frac{V^{\sigma,m}}{V^{\sigma,s}} = \frac{V^{\varphi,m}}{V^{\varphi,s}}. \quad (3)$$

Equation (3) is the key equilibrium condition. The terms of marriage must make it equally attractive relative to singlehood on the two sides of the market. The transfer adjusts to achieve this balance, taking account of the fertility response it induces. Together,  $(\alpha, n^m)$  describes the marital bundle: the transfer clears the market, and the wife's household choice determines fertility conditional on that transfer.

The competitive interpretation follows [Choo and Siow \(2006\)](#): individuals are atomistic and take the terms of marriage as given. Here, changing  $\alpha$  changes the bundle itself. A larger transfer changes both the division of household resources and the number of children, so both spouses' valuations respond to the wife's fertility choice.

### Aggregate outcomes

The three outcomes of interest are aggregate fertility, marriage, and the gender income gap.<sup>8</sup> Their connection through female labor supply is

$$\begin{aligned} n &= \mathcal{M}n^m + (1 - \mathcal{M})n^s, \\ l^{\varphi} &= 1 - \chi n, \\ \Gamma^y &= \frac{y^{\sigma}}{y^{\varphi}} = \frac{\Gamma^w}{l^{\varphi}}. \end{aligned} \quad (4)$$

Fertility is measured per woman. Labor incomes are measured before private transfers:  $y^{\sigma} = w^{\sigma}$  and  $y^{\varphi} = w^{\varphi}l^{\varphi}$ .

The last relation captures the distinction between wages and incomes. Even at a fixed wage gap, the income gap widens when women spend more time outside the market caring for children. Fertility therefore affects gender income inequality through labor supply, while marriage affects

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<sup>8</sup>Marriage is the model's theoretical family-structure outcome, while the preferred empirical measure is the share of children living with both parents. With relative marital fertility defined as  $q = n^m/n^s$ , that share equals

$$\frac{\mathcal{M}q}{1 + \mathcal{M}(q - 1)}.$$

It is strictly increasing in both  $\mathcal{M}$  and  $q$ . The two family-structure measures consequently move in the same direction in the individual-policy comparisons below and in Proposition 4. This monotonicity gives their comparative statics a common interpretation. Their levels remain distinct, and their movement under other changes depends on fertility composition. Appendix C.4 provides the mapping.

fertility by changing household resources and the composition of families.

### 3.2 Equilibrium Characterization

Before solving the model, I state the equilibrium formally.

**Definition 1** (Competitive equilibrium). *Given primitives  $(\beta, \rho, \chi, \lambda, \theta)$  and wages  $(w^{\sigma}, w^{\varphi})$ , a competitive equilibrium consists of single women's choices, married women's choices, a transfer share  $\alpha$ , and marriage rates satisfying:*

1. *Single women optimize subject to their resource and time constraints. Single men remain childless and consume their earnings.*
2. *Wives optimize conditional on the transfer and wages.*
3. *Husbands evaluate marriage at the fertility chosen by wives.*
4. *Taste-adjusted choices generate the participation rates in (2).*
5. *The transfer clears the marriage market as in (3).*

The equilibrium can be solved in closed form. To keep the expressions manageable, define

$$\begin{aligned} r &\equiv \frac{\rho - 1}{\rho}, \\ C &\equiv \left( \frac{(1 - \beta)\chi}{\beta} \right)^{\rho} (w^{\varphi})^{\rho-1} + \chi, \\ K &\equiv 1 - \frac{\beta}{(1 - \beta)(Cw^{\sigma})^r}. \end{aligned} \tag{5}$$

The wife's first-order condition then gives

$$n^s = \frac{1}{C}, \quad n^m = \frac{1 + \alpha\Gamma^w}{C}. \tag{6}$$

Equation (6) makes the income-sharing mechanism transparent. Single and married women face the same monetary price of a child,  $w^{\varphi}\chi$ . Marriage adds the transfer while leaving that price unchanged. Marital fertility is consequently single fertility multiplied by the increase in resources relative to the wife's own earning capacity. If the transfer were zero, the two fertilities would coincide.

The closed-form solution must also respect the household time constraints. I therefore impose the following restrictions.

**Assumption 1** (Interior baseline equilibrium). *The primitives satisfy*

$$Cw^{\sigma^*} > \left( \frac{\beta}{1-\beta} \right)^{1/r}, \quad \frac{\chi(1+\Gamma^w)}{C(1+K^{1/r}\Gamma^w)} < 1. \quad (7)$$

*Policy comparisons remain within the region with strictly positive consumption and market labor supply.*

The first inequality implies  $0 < K < 1$  and delivers strictly positive consumption after the transfer. The second ensures that the associated marital fertility leaves the wife non-negative time for market work. Together, they ensure that the marriage-market solution corresponds to a feasible interior household allocation. Single women's labor supply is already positive because  $C > \chi$ .

**Lemma 1** (Baseline equilibrium). *Under Assumption 1, the interior equilibrium is unique and satisfies*

$$\alpha^* = \frac{1 - K^{1/r}}{1 + K^{1/r}\Gamma^w} \in (0, 1), \quad n^m = \frac{1 + \Gamma^w}{C(1 + K^{1/r}\Gamma^w)}. \quad (8)$$

*The transfer and household fertilities are independent of  $\lambda$  and  $\theta$ .*

The proof appears in Appendix C.3. The independence from  $\lambda$  has a useful interpretation. A common proportional increase in the benefit of marriage makes marriage more attractive to both genders. It raises participation while preserving the transfer needed to balance the two sides, because the common benefit cancels from (3). Similarly,  $\theta$  governs participation's responsiveness to gains, while the equality of those gains determines the equilibrium transfer.

**Lemma 2** (Marriage gains). *In the baseline equilibrium,*

$$\begin{aligned} \frac{V_{\varphi,s}^{\varphi,m}}{V_{\varphi,s}^{\varphi,s}} &= \lambda(1 + \alpha^*\Gamma^w), & \frac{n^m}{n^s} &= 1 + \alpha^*\Gamma^w > 1, \\ \mathcal{M} &= \frac{[\lambda(1 + \alpha^*\Gamma^w)]^\theta}{1 + [\lambda(1 + \alpha^*\Gamma^w)]^\theta}. \end{aligned} \quad (9)$$

Lemma 2 states the economic logic compactly. Women's gains from marriage have a common marriage-benefit component,  $\lambda$ , and an income-sharing component,  $1 + \alpha^*\Gamma^w$ . The latter depends on what the transfer is worth relative to the wife's own earning capacity. Because the price of fertility is unchanged, the same factor scales consumption, fertility, and utility across being single versus being married. Market clearing equates the spouses' marriage-to-singlehood utility ratios, while their utility levels can differ. Appendix C.2 provides the proof.<sup>9</sup>

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<sup>9</sup>For completeness, Appendix C.1 also writes the husband's participation condition as a transfer schedule conditional on marriage and fertility. That schedule shows how much he is willing to transfer for a given marital bundle. It complements the wife's fertility first-order condition: she chooses fertility, and he decides whether the resulting contract is worth entering.

### 3.3 The Three-Way Trade-Off

Equations (4) and (9) bring the three outcomes together. Marriage is more attractive when income sharing adds substantially to the wife's resources. A wider wage gap can support that attraction, while also widening the gender income gap. Higher fertility adds a further tension by reducing the female market work that could otherwise offset the wage disparity.

The three possible pairs of objectives reveal different sides of the same problem:

- (1) **High fertility and widespread marriage.** Higher fertility leaves women less time for market work. If marriage is sustained by a substantial male wage advantage and the transfers it supports, the resulting labor-supply difference compounds the gender income gap.
- (2) **High fertility and gender income equality.** When women spend substantial time raising children, income equality requires a relatively small male wage advantage. This leaves less scope for transfers to make marriage attractive relative to women's own earning opportunities.
- (3) **Widespread marriage and gender income equality.** At a given common benefit of marriage, sustaining participation requires an attractive income-sharing component. Reconciling that component with a small income gap is easier when women supply more market labor, which in the baseline requires devoting less time to children.

The scope of this tension depends on the economic environment. The transfer bound makes that dependence precise: every interior baseline equilibrium satisfies

$$\frac{1}{\lambda} \left( \frac{\mathcal{M}}{1 - \mathcal{M}} \right)^{1/\theta} < 1 + \Gamma^y(1 - \chi n). \quad (10)$$

At fixed  $\lambda$  and  $\chi$ , sufficiently ambitious joint targets violate (10). Raising the marriage target increases the left side, while raising fertility or lowering the income-gap target reduces the right side. Appendix C.8 derives the restriction.

The policy question is how the available instruments move the three outcomes within this economic environment. The next subsection establishes why the conventional instruments considered individually leave the quandary unresolved.

### 3.4 Policy Remedies

I consider three familiar policy objectives in turn: making marriage more attractive, improving women's wage opportunities, and reducing the time cost of children. Each experiment changes one primitive while holding the others fixed. These experiments isolate mechanisms. Applying them to a complete fiscal program would additionally require specifying financing and the other features of the reform.

## Pro-marriage policies

Marriage bonuses, joint taxation, and related measures seek to make marriage more attractive, partly because of its connection to family stability and children's outcomes (Kearney, 2023; Autor et al., 2019). I represent their common-benefit component by an increase in  $\lambda$ .

**Proposition 1.** *An increase in  $\lambda$  raises aggregate fertility  $n$  and marriage  $\mathcal{M}$  but widens the gender income gap  $\Gamma^y$ .*

The logic is compositional. A higher  $\lambda$  attracts more people into marriage while leaving the equilibrium transfer and fertility within each marital state unchanged. Since married women have more children than single women, average fertility rises as more women marry. Those additional children require maternal time. Female labor supply falls, and the income gap widens. The policy succeeds on marriage and fertility through a channel that works against income equality. Appendix C.5 proves the result.

## Gender equality policies

Anti-discrimination and pay-equity measures seek to improve women's labor-market opportunities and reduce misallocation (Hsieh et al., 2019). I isolate their wage effect through an increase in  $w^{\text{f}}$  at fixed  $w^{\text{m}}$ .

**Proposition 2.** *An increase in  $w^{\text{f}}$ , holding  $w^{\text{m}}$  fixed, reduces aggregate fertility  $n$  and marriage  $\mathcal{M}$  while narrowing the gender income gap  $\Gamma^y$ .*

Higher female wages act through two channels. First, children become more expensive in terms of forgone earnings, reducing fertility among both single and married women. Second, a husband's transfer adds less relative to the wife's own earning capacity, reducing the income-sharing attraction of marriage. Fewer marriages then reinforce the fertility decline.

The transfer share responds to competing forces. Wage convergence and the changing value of fertility exert opposing effects on  $\alpha^*$ . Their combined effect unambiguously reduces the transfer relative to women's own earning capacity,  $\alpha^*\Gamma^w$ , and therefore the common marriage premium. Meanwhile, women both earn more per unit of work and spend more time in the market, so the income gap narrows through wages and labor supply together. Appendix C.6 establishes these claims.

## Reductions in childcare time costs

A lower time requirement per child appears to offer a direct way to support fertility while preserving women's market work. To examine that possibility, I reduce  $\chi$  while holding wages and the remaining primitives fixed.

**Proposition 3.** *A reduction in  $\chi$  raises aggregate fertility  $n$  and marriage  $\mathcal{M}$  but widens the gender income gap  $\Gamma^y$ .*

Lower childcare costs raise fertility in both marital states. They also increase the equilibrium value of the marital bundle, supporting a larger transfer and more marriage. These direct and composition effects both raise aggregate fertility.

Why, then, does female labor supply fall? The answer lies in the distinction between care per child and total care. With  $\rho > 1$ , families respond sufficiently strongly to the lower time price that  $\chi n$  rises even though  $\chi$  falls. Mothers spend less time on each child but more time on childcare altogether. The resulting reduction in market work widens the income gap. Appendix C.7 establishes the response of total childcare time.

### Three instruments, three incomplete solutions

Table 4 summarizes the results. Each conventional instrument succeeds on part of the agenda while leaving at least one objective moving in the wrong direction. Pro-marriage and lower-childcare-cost policies raise fertility and marriage at the expense of income equality. Higher female wages narrow the income gap but reduce fertility and marriage.

| Policy                                       | Fertility $n$ | Marriage $\mathcal{M}$ | Income gap $\Gamma^y$ |
|----------------------------------------------|---------------|------------------------|-----------------------|
| Pro-marriage ( $\uparrow \lambda$ )          | +             | +                      | + (widens)            |
| Gender equality ( $\uparrow w^\varnothing$ ) | −             | −                      | − (narrows)           |
| Lower childcare costs ( $\downarrow \chi$ )  | +             | +                      | + (widens)            |

Table 4. Effects of Conventional Policies Considered Individually

*Notes:* Each row changes one instrument while holding the remaining primitives fixed. Comparisons remain within the interior equilibrium region.

Coordinating instruments creates additional possibilities. Appendix C.8 shows that a sufficiently small increase in female wages combined with an appropriately chosen increase in the marriage benefit can raise fertility and marriage while narrowing the gender income gap. However, such an approach requires delicate balancing and coordination that is hard to achieve in reality. Therefore, could there be a simpler policy approach that relaxes the three-way tension in the quandary? The next section provides an answer.

## 4 Redistributing Childcare Between Spouses

The baseline assigns all childcare to mothers. Time-use evidence documents a substantial maternal share of care (Aguiar and Hurst, 2007; Kleven et al., 2019), and institutions and policies, including

reserved paternity leave, can change fathers' involvement (Farré and González, 2019). This raises a natural question: can sharing care more equally support fertility and marriage while maintaining or improving gender income equality?

Answering that question requires following the time savings through both spouses' budgets. When the husband provides additional care, he earns less, and some of that income would have financed his transfer to the wife. He must also be willing to accept the new balance between consumption and children. To capture these effects, I extend the model to allow both spouses to provide childcare and let the transfer adjust to clear the marriage market.

#### 4.1 Setup and Household Choices

Let  $\chi^{\varnothing}$  and  $\chi^{\sigma}$  denote the time requirements per child borne by wives and husbands:

$$\chi^{\varnothing} > \chi^{\sigma} \geq 0, \quad \chi^{\varnothing} + \chi^{\sigma} = \chi. \quad (11)$$

Single mothers continue to provide  $\chi$  units of care per child. Marriage thus changes who provides the care while preserving the total parental time required for a child. Appendix D.7 gives the corresponding analysis when total requirements differ by marital status.

The wife's budget becomes

$$c^{\varnothing,m} = \alpha w^{\sigma} (1 - \chi^{\sigma} n^m) + w^{\varnothing} (1 - \chi^{\varnothing} n^m),$$

and the husband's budget becomes

$$c^{\sigma,m} = (1 - \alpha) w^{\sigma} (1 - \chi^{\sigma} n^m).$$

The transfer remains a share of the husband's earnings, which now depend on his childcare obligations. As before, the wife chooses fertility conditional on the transfer share, and the husband evaluates the resulting bundle.

It is useful to rewrite the wife's budget in terms of effective resources and an effective childcare cost:

$$\tilde{w}^{\varnothing} = w^{\varnothing} + \alpha w^{\sigma}, \quad \tilde{\chi}^{\varnothing} = \frac{w^{\varnothing} \chi^{\varnothing} + \alpha w^{\sigma} \chi^{\sigma}}{w^{\varnothing} + \alpha w^{\sigma}}. \quad (12)$$

Her budget is then  $c^{\varnothing,m} = \tilde{w}^{\varnothing} (1 - \tilde{\chi}^{\varnothing} n^m)$ . The effective childcare cost is an income-weighted average: a child uses some of her own earning capacity and some of the husband's earning capacity that finances her transfer.

The interior fertility choices are

$$n^m = \frac{1}{\left(\frac{(1-\beta)\tilde{\chi}^\varnothing}{\beta}\right)^\rho (\tilde{w}^\varnothing)^{\rho-1} + \tilde{\chi}^\varnothing}, \quad n^s = \frac{1}{C}. \quad (13)$$

This is the same household trade-off as before, expressed using the resources and fertility price relevant within marriage.

**Lemma 3** (Indirect utility with shared childcare). *The wife's marital value is*

$$V^{\varnothing,m} = \lambda(1-\beta)^{1/r} \tilde{w}^\varnothing \left[ 1 + \left(\frac{\beta}{1-\beta}\right)^\rho (\tilde{w}^\varnothing \tilde{\chi}^\varnothing)^{1-\rho} \right]^{1/(\rho-1)}. \quad (14)$$

Her single value follows by setting  $\lambda = 1$ ,  $\tilde{w}^\varnothing = w^\varnothing$ , and  $\tilde{\chi}^\varnothing = \chi$ .

Equations (13) and (14) show why the effective price matters. At fixed resources, a lower price raises fertility and the value of marriage to the wife. With  $\rho > 1$ , she substitutes sufficiently strongly toward children that consumption falls. Her utility gain comes from the opportunity to choose a more attractive combination of consumption and fertility. Appendix D.1 provides the derivation.

## 4.2 When Redistribution Improves All Three Outcomes

Consider a local policy that shifts one unit of childcare time per child from the wife to the husband:

$$\frac{d\chi^\sigma}{d\delta} = 1, \quad \frac{d\chi^\varnothing}{d\delta} = -1.$$

Wages, preferences, and single mothers' childcare costs remain fixed. Total derivatives below include the equilibrium adjustment of  $\alpha$ ; at  $\chi^\sigma = 0$ , they are interpreted as right-hand derivatives. Appendix D.2 derives the local equilibrium response formulas used below.

Three sufficient conditions organize the result. The first and second concern the female side of the reform: does it make fertility less costly for the wife, and does it improve women's earnings relative to men's after fertility adjusts? The third asks whether the husband is willing to enter the modified contract. Together, they follow the change in caregiving through household choices, marriage participation, and relative earnings.

**Condition 1: the wife faces a lower effective childcare cost.** At the initial transfer, redistribution lowers the wife's effective childcare cost if

$$w^\varnothing > \alpha w^\sigma. \quad (15)$$

The intuition is straightforward. An hour released from the wife's childcare raises her potential earnings by  $w^{\varnothing}$ . An additional hour of the husband's childcare reduces the earnings available to finance her transfer by  $\alpha w^{\sigma}$ . If the first exceeds the second, shifting care toward the husband lowers the price of children from her perspective. Her desired fertility and marital value both rise at the initial transfer.

Condition (15) compares the earning capacities affected by an additional hour of care. Realized earnings and transfers also depend on fertility, so the condition is stated in terms of wages and the transfer share.

**Condition 2: relative earnings converge after fertility adjusts.** The second condition follows the reform through aggregate childcare. If redistribution raises fertility and marriage, it also creates additional care to be provided. Income convergence depends on how that larger total is divided between mothers and fathers.

Average labor supplies and the income ratio are

$$\begin{aligned} l^{\sigma} &= 1 - \chi^{\sigma} \mathcal{M}n^m, \\ l^{\varnothing} &= 1 - \chi^n + \chi^{\sigma} \mathcal{M}n^m, \\ \Gamma^y &= \Gamma^w \frac{l^{\sigma}}{l^{\varnothing}}. \end{aligned} \tag{16}$$

The income-gap condition is

$$\chi \frac{dn}{d\delta} < \left[ \mathcal{M}n^m + \chi^{\sigma} \frac{d(\mathcal{M}n^m)}{d\delta} \right] \left( 1 + \frac{l^{\varnothing}}{l^{\sigma}} \right). \tag{17}$$

The left side is the increase in total childcare generated by higher fertility. The bracket on the right is the increase in fathers' aggregate childcare, including their care for additional marital births. Moving an hour of care from mothers to fathers affects both sides of the income ratio: it releases female market time and reduces male market time. The final factor accounts for this two-sided effect.

Condition (17) requires women's earnings to improve relative to men's. This can occur even if the fertility response reduces female labor supply, provided male labor supply falls proportionately more.

**Condition 3: the husband gains from the adjusted marital contract.** The equilibrium response also depends on the husband's valuation of the reform. He loses earnings when he provides more care, but he may value the additional fertility enough to accept that loss. Moreover, the transfer can

adjust to redistribute the gains. To express this trade-off, define

$$z \equiv \tilde{\chi}^{\varnothing} n^m \in (0, 1), \quad R \equiv \frac{\beta}{1 - \beta} \left( \frac{c^{\sigma, m}}{n^m} \right)^{1/\rho} - (1 - \alpha) w^{\sigma} \chi^{\sigma}.$$

Here  $z$  is the share of the wife's effective resources devoted to fertility. The term  $R$  is the husband's marginal value of another child, measured in consumption units and net of the earnings cost of his existing care obligation.

The male-side condition is

$$\frac{R(w^{\varnothing} - \alpha w^{\sigma}) \rho (1 - z)}{\tilde{w}^{\varnothing} \tilde{\chi}^{\varnothing}} > (1 - \chi^{\sigma} n^m) (w^{\sigma} - w^{\varnothing}). \quad (18)$$

A compensated comparison gives (18) its economic meaning. Imagine adjusting the transfer as childcare is reallocated so that the wife's gain from marriage stays at its original level. Fertility responds to both the lower childcare cost and this transfer adjustment. The condition requires the husband to gain along that path. After ensuring that the wife is no worse off, the additional children must be valuable enough to leave a gain for him as well.

The left side measures the husband's net valuation of that induced fertility response. The right side captures the earnings cost of shifting care toward the higher-wage spouse, after accounting for the compensating transfer. If the inequality holds, the reform creates room to make marriage more attractive to both genders. The actual market-clearing transfer then shares that improvement, raising both spouses' marriage premiums. Appendix D.3 derives this interpretation.

**Proposition 4** (Redistributing childcare). *Consider an interior equilibrium with  $w^{\sigma} > w^{\varnothing} > 0$ ,  $0 < \alpha < 1$ , and (11). If Conditions (15) and (18) hold, the equilibrium is locally regular and a sufficiently small redistribution toward fathers raises marriage, marital fertility, and aggregate fertility. Both spouses' marriage-to-singlehood utility ratios rise. If Condition (17) also holds, the gender income gap narrows. Thus the three conditions together imply*

$$\frac{dn}{d\delta} > 0, \quad \frac{d\mathcal{M}}{d\delta} > 0, \quad \frac{d\Gamma^y}{d\delta} < 0.$$

The proof appears in Appendix D.4. The result works through household choices and market participation together. The lower effective price encourages fertility, while the husband's gain from the adjusted bundle supports a higher common marriage premium. More people marry, and married women choose more children. Because single fertility is unchanged and marital fertility is higher, both changes raise aggregate fertility.

Income convergence requires a further step. Additional children bring additional care obli-

gations, and redistribution must be sufficient to improve women's earnings relative to men's after those obligations are taken into account. Conditions (15) and (18) resolve the participation problem; Condition (17) resolves the relative-income problem.

Starting from the baseline,  $\chi^{\sigma} = 0$ , the third condition simplifies to

$$\chi \frac{dn}{d\delta} < \mathcal{M}n^m(1 + l^{\varphi}).$$

Appendix D.5 gives the corresponding response formulas using the closed-form baseline equilibrium.

The broader lesson is bilateral. Mothers' cost of children and fathers' valuation of the resulting marital bundle jointly determine how fertility responds. Redistribution can improve all three outcomes when it creates an attractive exchange for both sides of the marriage market and leaves relative earnings more equal after families adjust.

### 4.3 When the Conditions Fail

The same framework clarifies why a reform can help in one setting and disappoint in another. For example, the equilibrium transfer response can offset a direct reduction in the wife's childcare cost. Alternatively, a contract that raises both spouses' marriage gains can also generate enough additional care to prevent income convergence. These are distinct ways for joint improvement to fail.

For the following proposition, maintain  $w^{\sigma} > w^{\varphi} > 0$ ,  $0 < \alpha < 1$ , interior household allocations, and the fixed-total-care experiment. Consider a regular equilibrium branch on which

$$\frac{\partial}{\partial \alpha} \log \left[ \frac{V^{\varphi,m}/V^{\varphi,s}}{V^{\sigma,m}/V^{\sigma,s}} \right] > 0.$$

This condition says that increasing the transfer raises women's willingness to marry relative to men's, including the induced fertility response. Conditions (15) and (18) imply it.

**Proposition 5** (Alternative responses). *At an equilibrium satisfying the preceding regularity condition:*

- (A) *If  $w^{\varphi} \leq \alpha w^{\sigma}$ , the wife's fertility and marital utility weakly fall at the initial transfer. Starting from  $\chi^{\sigma} = 0$ , marriage, marital fertility, and aggregate fertility fall in equilibrium, while the gender income gap narrows.*
- (B) *If  $w^{\varphi} > \alpha w^{\sigma}$  and the husband's marital utility weakly falls at the initial transfer, the equilibrium transfer falls. The equilibrium marriage and fertility responses depend on the transfer-adjusted incentives.*

*(C) If Conditions (15) and (18) hold but Condition (17) fails, marriage and fertility rise, while the gender income gap is locally unchanged or widens.*

Appendix D.6 provides the proof. Case (A) reverses the wife’s price mechanism. When her budget depends sufficiently heavily on her husband’s earnings, shifting care toward him removes more transfer-financing capacity than it releases own earning capacity. At the baseline, marriage and fertility decline. With positive paternal care initially, additional interactions arise because lower fertility also reduces the husband’s existing childcare burden.

Case (B) illustrates the importance of the transfer adjustment. At the original transfer, the wife gains and the husband’s utility weakly falls. The transfer must fall to restore balance. That adjustment returns some resources to the husband while changing the wife’s fertility choice, so the equilibrium fertility response can differ from its initial direction.

Case (C) concerns aggregate time allocation. The marriage-market response is favorable: both sides value the contract more, and fertility rises. The additional childcare is sufficiently large, however, that the per-child redistribution fails to narrow the aggregate income gap. Relative earnings depend on the division of care across all children as well as the division per child.

The relative strength of these mechanisms depends on equilibrium quantities that change with wages, preferences, and childcare arrangements. Their implications for high-income or middle-income countries therefore require evaluation at each economy’s starting point which call for careful estimation of parameters.

#### **4.4 Mapping to the Empirical Evidence**

Reserved paternity leave provides a natural setting in which to study changes in fathers’ caregiving. Evidence from Spain (Farré and González, 2019), Norway (Cools et al., 2015), Denmark (Kleven et al., 2026), South Korea (Lee and Lee, 2025), and Quebec (Patnaik, 2019) examines how reforms change the household division of labor. These outcomes address the first practical question: does policy move childcare toward fathers? The literature generally finds a positive answer.

The next question is whether that change supports fertility and marriage. The literature reports differing fertility responses across the settings studied. For example, Farré and González (2019) finds a reduction in subsequent fertility while Lee and Lee (2025) finds an increase in fertility following paternal leave policies. The model offers a way to organize this heterogeneity. A reform changes mothers’ time constraints alongside fathers’ consumption, private transfers, and the attraction of the marital bundle. Similar shifts in caregiving can therefore produce different fertility responses depending on preferences and initial conditions. This calls for a case-by-case calibration à la Section 5.

The third question is whether the gender income gap narrows after families respond. Higher

maternal employment is one route to income convergence. The relative-income measure used in this paper also captures changes on the male side, allowing convergence when men's earnings fall proportionately more. Evaluating the full prediction therefore calls for information on wages, labor supply, fertility by marital status, private transfers, and marriage participation.

#### **4.5 Public Childcare and the Marketization of Childcare**

Public childcare and purchased care provide another way to release parental time. Outsourcing can reduce total parental care per child, introduce a monetary payment, and change the two parents' remaining obligations differently. Its effects therefore combine several mechanisms.

The baseline reduction in  $\chi$  isolates a lower maternal time requirement while holding additional spending and financing outside the experiment. Proposition 3 shows that, under  $\rho > 1$ , the fertility response is strong enough for total maternal childcare to rise. This result highlights the distinction between time saved per child and total time saved after fertility adjusts. Applying it to public childcare would require accounting for the reform's other effects on household resources and choices.

The redistribution experiment focuses on who provides the remaining care. Moving an hour from the wife to the husband lowers his earnings as it releases hers. Purchased childcare replaces parental time with a service whose cost must be financed. The associated budgets and participation incentives differ, so extending Proposition 4 to outsourced care requires incorporating those payments and their incidence.

The common lesson is to follow the reform through the household and the marriage market. How does it change the wife's effective price of children? What contract is the husband willing to accept after fertility and transfers adjust? And how is total childcare ultimately divided between parents and the market? A model with purchased childcare would answer these questions by specifying prices, financing, and substitution with parental time.

### **5 Quantitative Assessment**

I calibrate the model to Mexico from 1990 to 2015. The exercise uses its long time series, substantial productivity and wage changes, and reversal of the gender education gap. The empirical family-structure target is children's coresidence with both parents, mapped from the model's marriage rate and relative fertility.

## 5.1 Data and Calibration

Figure 4 plots the transition used in the calibration. Fertility falls from approximately 3.5 to 2.1 children per woman. The share of children living with both parents falls from 81 to 75 percent, while the share living only with their mother rises from 11 to 19 percent. Women’s labor income share rises from 24 to 32 percent. Real GDP per capita increases from roughly \$12,000 to over \$18,000 in 2017 dollars. Women aged 25 to 35 move from 0.8 fewer years of schooling than men in 1990 to a small advantage by 2015.

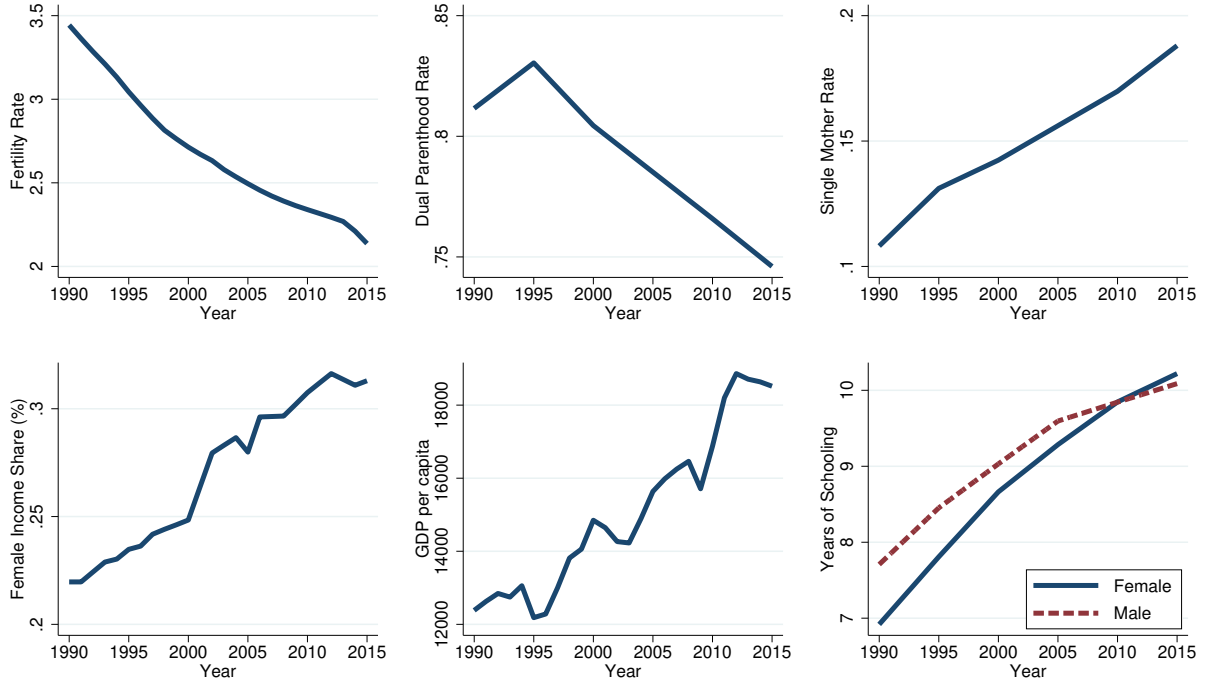


Figure 4. Mexico’s Transition Path (1990–2015)

*Notes:* Fertility is children per woman, from the United Nations. Family structure measures children’s residence with both parents or only their mother (Brenøe and Wasserman, 2025). Female labor income shares are from the World Inequality Database. GDP per capita is in 2017 U.S. dollars, from the OECD. Education gaps use average years of schooling at ages 25–35 (Barro and Lee, 2013).

Time-varying wages are the sole exogenous drivers. I decompose them into gender-neutral productivity  $A_t$ , the gender human-capital gap  $\Gamma_t^h$ , and residual gender-biased productivity  $B_t$ :

$$w_t^{\sigma} = A_t, \quad w_t^{\varnothing} = \frac{A_t}{B_t \Gamma_t^h}. \quad (19)$$

Thus  $B_t = (w_t^{\sigma} / w_t^{\varnothing}) / \Gamma_t^h$ .

The procedure has three steps. First, I set  $\chi = 0.075$  following De La Croix and Doepke (2003) and construct relative human capital from schooling using Caselli (2005). Second, I infer

female labor supply from observed fertility using (4) and recover wages from GDP per capita and female labor income shares. Third, I choose  $\{\beta, \rho, \lambda, \theta\}$  to match fertility and the child-based family-structure series.

The parameters govern distinct margins:  $\beta$  primarily controls fertility’s level,  $\rho$  its response to productivity,  $\lambda$  the level of marriage and thus the mapped family-structure measure, and  $\theta$  participation responsiveness. These roles guide the matching exercise. Formal identification would require a separate analysis of how the moments jointly distinguish the parameters. Table 5 reports the estimates.

|              | Interpretation                         | Value | Target                                        | Source                                      |
|--------------|----------------------------------------|-------|-----------------------------------------------|---------------------------------------------|
| $\chi$       | Time cost of children                  | 0.075 | <a href="#">De La Croix and Doepke (2003)</a> |                                             |
| $A_t$        | Gender-neutral productivity            |       | GDP per capita                                | OECD                                        |
| $B_t$        | Gender-biased productivity             |       | Female labor income share                     | World Inequality Database                   |
| $\Gamma_t^h$ | Gender human-capital gap               |       | Years of schooling                            | <a href="#">Barro and Lee (2013)</a>        |
| $\beta$      | Fertility preference weight            | 0.104 | Fertility level                               | United Nations                              |
| $\rho$       | Consumption–fertility substitutability | 1.50  | Fertility trend                               | United Nations                              |
| $\lambda$    | Common marriage benefit                | 1.03  | Dual-parenthood level                         | <a href="#">Brenøe and Wasserman (2025)</a> |
| $\theta$     | Participation responsiveness           | 3.40  | Dual-parenthood trend                         | <a href="#">Brenøe and Wasserman (2025)</a> |

Table 5. Calibration Results

*Notes:* Time-varying inputs are constructed from data. Preference and marriage parameters minimize the distance between model and data for fertility and children’s coresidence with both parents over 1990–2015. The latter is mapped from marriage and relative fertility as described in Appendix C.4.

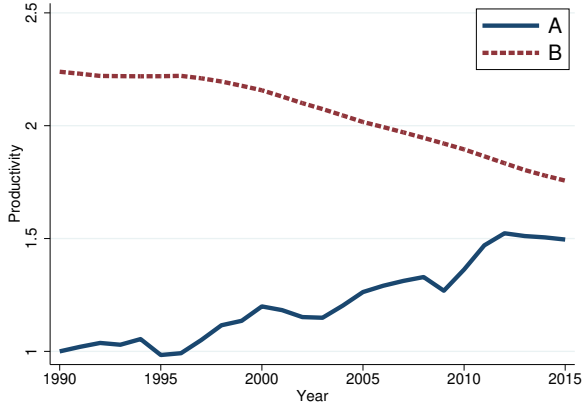
The estimate  $\rho = 1.50$  places the economy in the substitution-dominant region. The estimated marriage benefit is  $\lambda = 1.03$ . The separate restrictions in (7) govern the existence and feasibility of the interior transfer.

Figure 5 displays the driving forces and model fit. Human capital converges from 1990 onward; residual gender bias begins to decline in the late 1990s; and gender-neutral productivity grows after the 1994 Peso Crisis. The model tracks the reported fertility and family-structure paths with four estimated parameters.

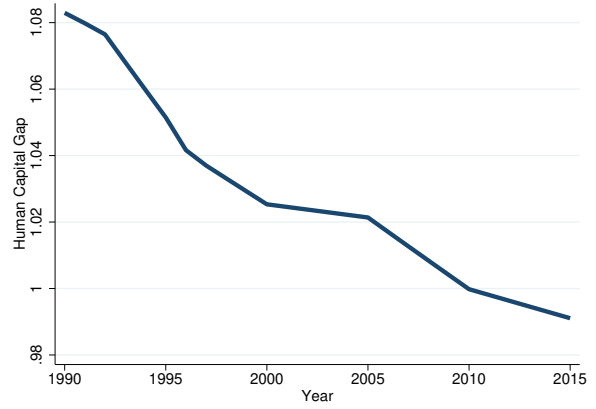
## 5.2 Equilibrium Outcomes

One advantage of the model is that it recovers intra-household transfers and gender-specific utility levels, objects that are difficult to observe directly. Figure 6 reports the key equilibrium quantities.

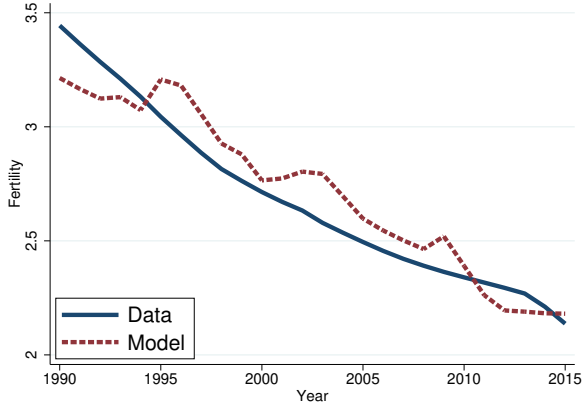
Figure 6a shows that the equilibrium transfer share  $\alpha_t$  falls by only about one percentage point despite the sharp decline in marital fertility. The intuition is that, as the wage gap narrows, husbands must transfer a larger share of a smaller relative income to sustain any given marital fertility through income effects, and these two forces roughly offset in the share.



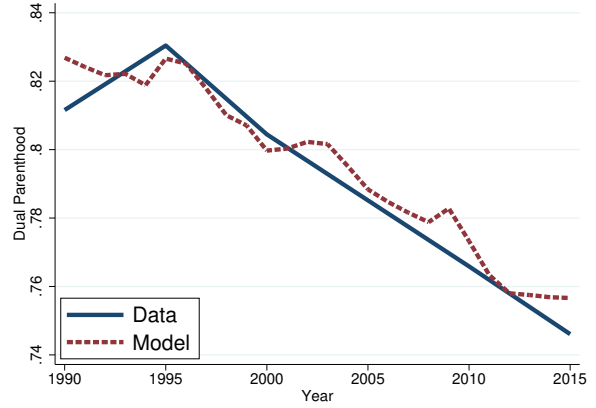
(a) Technologies ( $A_t, B_t$ )



(b) Human Capital Gap ( $\Gamma_t^h$ )



(c) Fertility



(d) Dual Parenthood

Figure 5. Model Fit: Mexico (1990–2015)

*Notes:* Solid lines denote data and dashed lines denote model values. Fertility is children per woman. Dual parenthood measures children's coresidence with both parents; the model maps adult marriage and relative fertility into this child-based measure. Productivity components are constructed using (19), GDP per capita, female labor income shares, and schooling. Parameters appear in Table 5.

Figure 6b documents how welfare realigns across family types. Single men gain ground relative to married men as marital fertility declines. Both single and married women gain substantially relative to married men as the wage gap closes. The husband-wife welfare gap within marriage shrinks from 42 to 26 percent, even though transfers decline in absolute share. Women's rising earning power, rather than larger transfers, drives this convergence.

Using the calibrated equilibrium, I evaluate the three conditions under which a small redistribution of childcare from wives to husbands raises fertility, marriage, and gender income equality. Because the baseline assigns all childcare to mothers, the simplified formulas in Appendix D.5 apply. Two conditions hold throughout 1990–2015: the wife faces a lower effective childcare cost at the initial transfer (Condition 15), and the relative-earnings condition remains satisfied after the

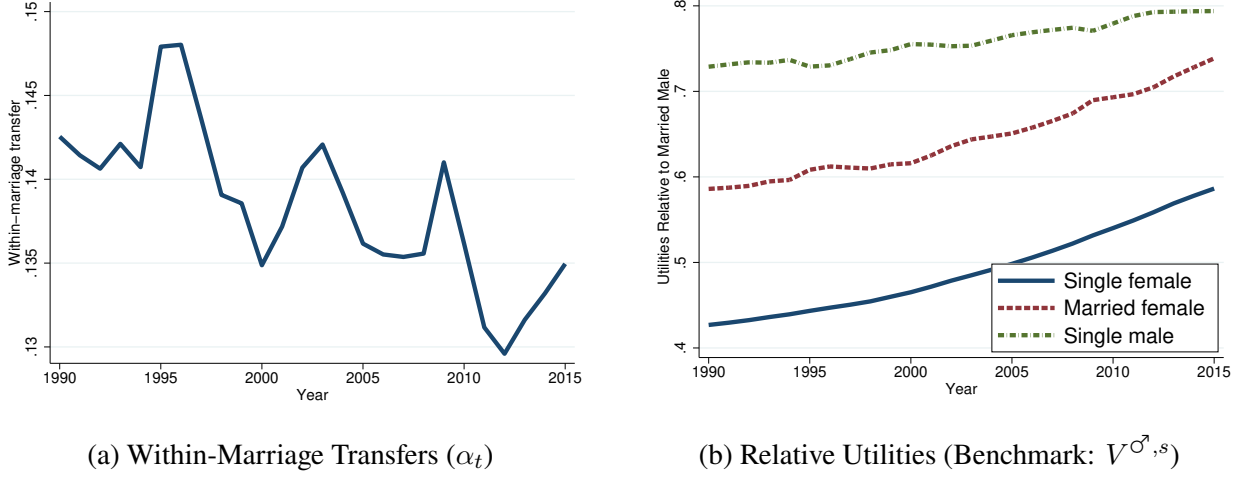


Figure 6. Model-Generated Equilibrium Outcomes

Notes: Simulations use Table 5. Panel (a) shows the transfer share. Panel (b) reports systematic utilities of married men, married women, and single women relative to single men; taste shocks are excluded.

induced fertility response (Condition 17). The male-side condition (Condition 18) is the binding margin in the early part of the sample. It fails from 1990 to 1997 and holds from 1998 onward, so the joint condition is satisfied in 18 of the 26 sample years. The timing follows the calibrated decline in residual gender-biased productivity and the associated narrowing of the spousal wage gap, which raises the value of marital fertility to husbands relative to the earnings cost of their additional childcare. By the end of the transition, the Mexican economy lies in the region where Proposition 4 predicts that childcare redistribution can simultaneously improve all three outcomes.

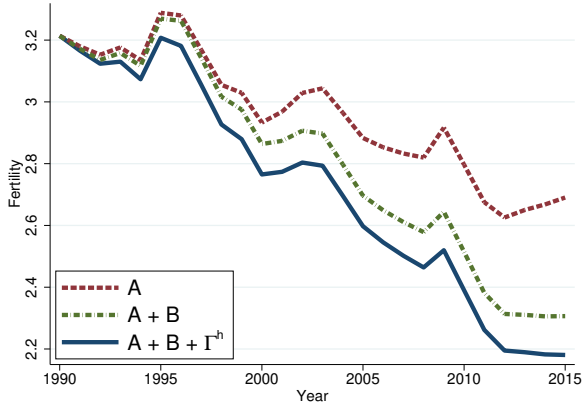
### 5.3 Decomposition

I decompose changes in fertility, the observed family-structure counterpart, the gender income gap, and relative spousal utility by sequentially activating  $A_t$ ,  $B_t$ , and  $\Gamma_t^h$ .<sup>10</sup> Figure 7 presents the results. As a sequential decomposition, the allocation of contributions depends on the stated ordering.

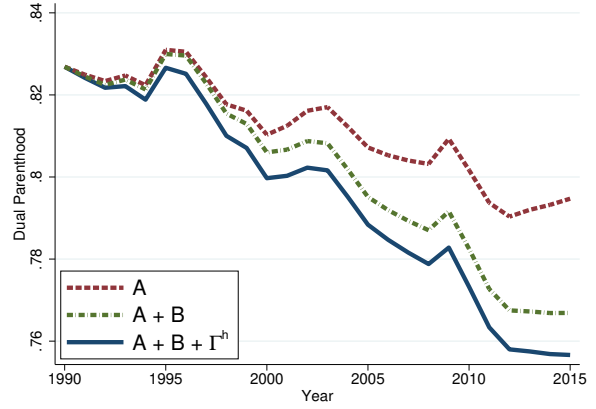
Gender-neutral productivity accounts for approximately half of the fertility decline and half of the deterioration in the child-based family-structure measure. Higher opportunity costs reduce fertility within marital states; lower marital fertility reduces marriage gains, reinforcing the aggregate decline. This driver contributes approximately 10 percent of income convergence through female labor supply and widens the systematic spousal utility gap as transfers fall.

Declining gender-biased productivity accounts for approximately 40 percent of the fertility and family-structure changes and most income and relative-utility convergence. At fixed male wages, lower  $B_t$  raises female wages. It reduces fertility through opportunity costs and lowers the relative

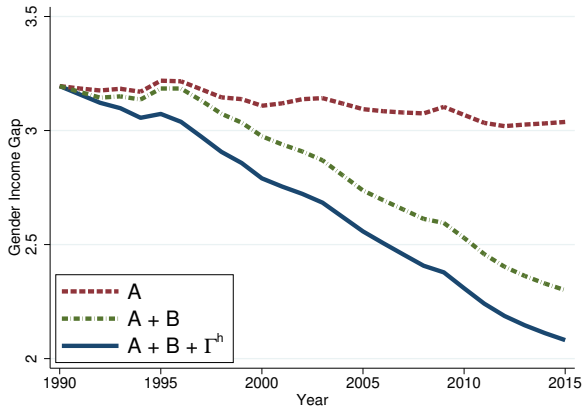
<sup>10</sup>Figure A.9 reports analogous decompositions for the other relative-utility measures.



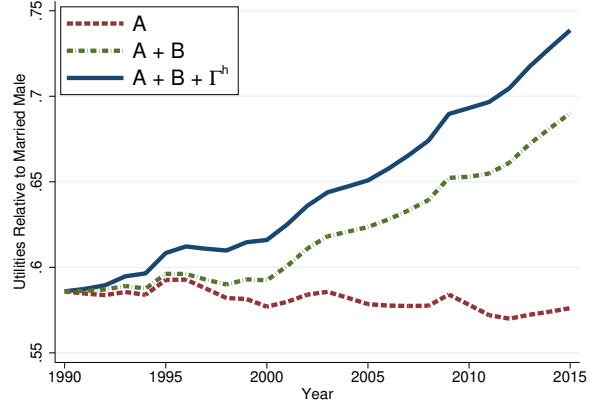
(a) Fertility



(b) Dual Parenthood



(c) Gender Income Gap



(d) Relative Utility:  $V^Q,m / V^{\sigma,m}$

Figure 7. Decomposition of Mexico's Transition (1990–2015)

Notes: “ $A_t$  only” varies gender-neutral productivity while fixing  $B_t$  and  $\Gamma_t^h$  at their 1990 values. “ $A_t + B_t$ ” additionally allows residual gender bias to vary. “All” is the full model. The family-structure panel uses the child-based counterpart of marriage.

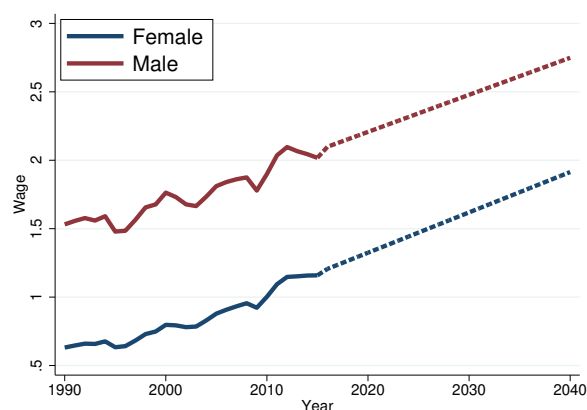
income-sharing gain from marriage.

Human-capital convergence accounts for the remainder. Because  $B_t$  is the residual wage gap after removing  $\Gamma_t^h$ , their relative contributions depend on the mapping from schooling into human capital.

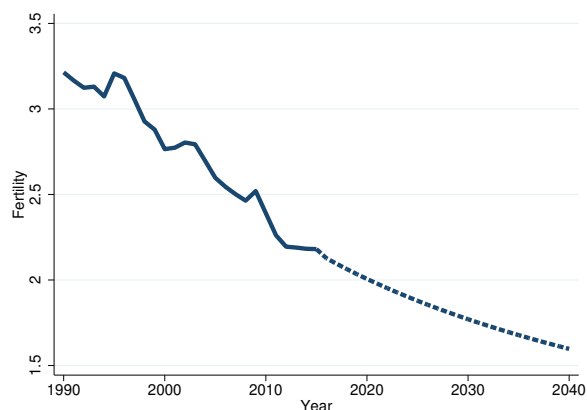
## 5.4 Prediction

I extrapolate wages linearly from their 1990–2015 trends and solve the model through 2040. Figure 8 reports fertility near 1.5 and a child-based family-structure share near 70 percent by 2040, together with continued income convergence. The reported 2023 fertility prediction is 1.93, compared with the data value of 1.91 used for the exercise. These projections are conditional on the extrapolated

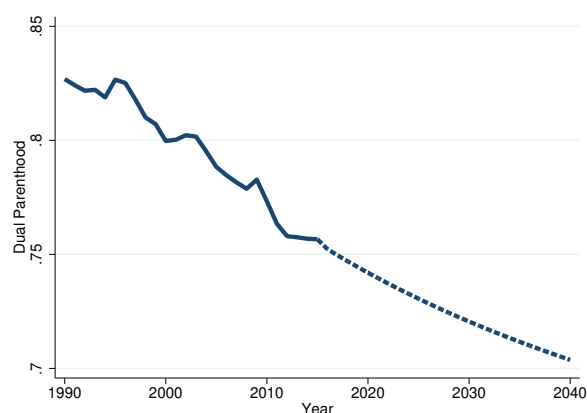
wage paths and maintained model parameters.



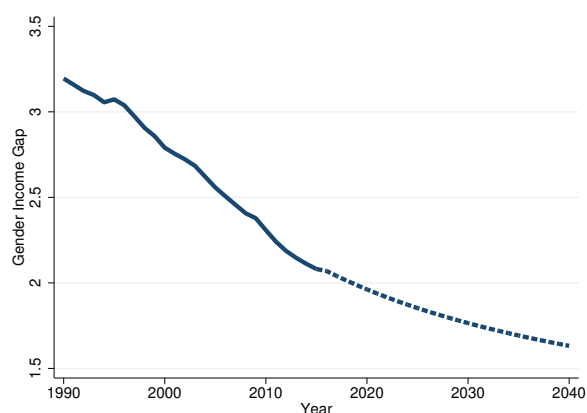
(a) Extrapolated Wages



(b) Fertility



(c) Dual Parenthood



(d) Gender Income Gap

Figure 8. Model-Based Predictions: Mexico to 2040

*Notes:* Wages are extrapolated from 1990–2015 trends and outcomes are generated using Table 5. The reported 2023 fertility prediction is 1.93 versus the United Nations data value of 1.91 used in the exercise. Family structure is children’s coresidence with both parents.

The value of the framework here is that it does not extrapolate fertility, marriage, and gender gaps independently. The three outcomes are treated as joint endogenous outcomes of gender-specific wage dynamics, which in turn are decomposed into three structurally interpretable drivers.

## 6 Dynamic Extension

The static model takes the gender human-capital gap as an input. Yet family structure may itself shape that gap in the next generation. A substantial literature links children’s outcomes to their family environment and finds that boys are often more sensitive than girls to disadvantage and the absence of a biological father (Sommers, 2001; Bertrand and Pan, 2013; Chetty et al., 2016;

Autor et al., 2019; Wasserman, 2020; Reeves, 2022; Frimmel et al., 2024). Autor et al. (2019) examine gender differences associated with family disadvantage, and Autor et al. (2023) study family structure and high-school performance.

These findings suggest a feedback worth incorporating. If lower marriage weakens future male human capital relative to female human capital, it changes the earning opportunities that support the next generation’s marital contracts. Today’s family choices can then affect tomorrow’s marriage market.

I introduce a reduced-form law of motion:

$$\Gamma_{t+1}^h = \mathcal{H}(\mathcal{M}_t), \quad \mathcal{H}'(\mathcal{M}_t) > 0. \quad (20)$$

Marriage serves as a summary predictor of the family environment. Equation (20) captures boys’ greater sensitivity to that environment, allowing the underlying mechanism to operate through parental inputs, role models, or peer and neighborhood effects. The relationship is imposed as a modeling assumption; establishing its causal magnitude and its connection to children’s living arrangements would require separate empirical analysis.

Figure 9 brings the channels together. The human-capital link runs across generations; time subscripts are omitted from the diagram to keep the mechanism visible.

The transition unfolds through two exogenous drivers and one endogenous feedback:

- (1) **Gender-neutral productivity growth** raises the opportunity cost of children. Fertility falls within both marital states, and the equilibrium transfer and income-sharing multiplier decline. Marriage becomes less attractive, so the shift toward singlehood reinforces lower aggregate fertility. Women devote less time to childcare, and the income gap narrows.
- (2) **Declining gender-biased productivity** improves women’s wages at a given male wage. Children become more costly in forgone female earnings, while the transfer adds less relative to women’s own resources. Fertility and marriage fall, and the income gap narrows through both wages and market work.
- (3) **Dynamic propagation** carries the marriage response into the next generation. Lower current marriage reduces future male relative human capital through (20). This weakens the income-sharing basis of future marriages and puts additional downward pressure on fertility.

Two implications stand out. First, both gender-neutral and gender-biased technological change can simultaneously depress fertility, erode marriage, and narrow gender income gaps because all three outcomes are tied together by the same equilibrium system. This complements earlier emphases on factor-biased technical change in Greenwood et al. (2005b) and Ngai and Petrongolo (2017). Second, policies that change marriage today may also change the distribution of earning

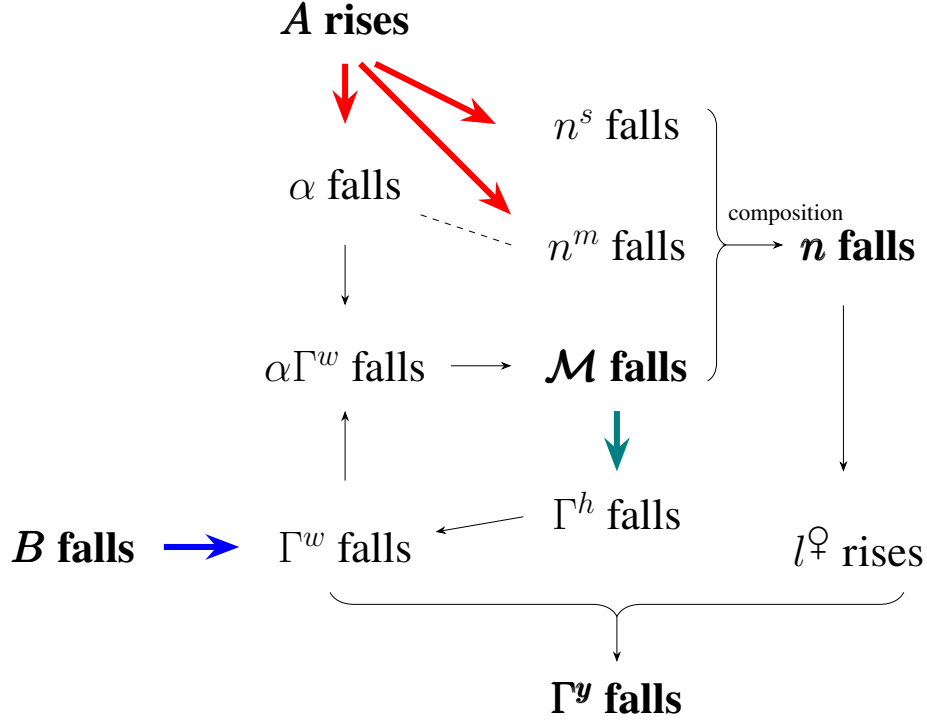


Figure 9. The Dynamic Mechanism

Notes: Red arrows represent the effects of rising gender-neutral productivity  $A_t$ . The blue arrow represents the effect of falling gender-biased productivity  $B_t$ . The teal arrow represents the dynamic human-capital effect through  $\mathcal{M}_t \rightarrow \Gamma_{t+1}^h$  in Equation (20). Time subscripts are suppressed. The diagram summarizes directions of propagation; their magnitudes and the stability of the system depend on the response elasticities.

opportunities tomorrow. Their long-run importance depends on the strength of the human-capital link together with the immediate demographic response.

## 7 Conclusion

This paper establishes a three-way trade-off: few societies simultaneously achieve high fertility, widespread dual parenthood, and gender income equality. Across OECD countries, a global panel of 95 economies, and U.S. states and commuting zones, the share of observations attaining all three is only 2.5–7.6 percent, far below the 12.5 percent implied by statistical independence. Mutual information analysis shows that the trade-off is genuinely three-dimensional: the variables exhibit synergy, so no combination of pairwise analyses can recover the joint structure. This statistical finding is what formally motivates studying fertility, marriage, and gender equality together rather than pair by pair.

A tractable marriage-market model explains the pattern. Marriage bundles an income transfer with shared fertility, so the market clears through an equilibrium contract involving both transfer and fertility. Because mothers provide most childcare, fertility is tied directly to female labor supply and

hence to the gender income gap. Three conventional instruments—pro-marriage benefits, higher female wages, and lower childcare costs—each improve two of the three outcomes at the expense of the third. Under these standard policies, the trade-off is binding.

The paper’s central theoretical contribution is to identify an instrument that can relax the trade-off: redistributing childcare from wives to husbands. The reform raises fertility, marriage, and gender income equality when it lowers the wife’s effective cost of children, preserves a gain for the husband after transfers adjust, and leaves relative earnings more equal once fertility responds. These conditions make the marital contract more attractive to both spouses, which is why the policy can succeed where one-sided interventions fail. Paternity-leave mandates with reserved father quotas are the natural policy counterpart.

The quantitative results reinforce the theory. The calibrated model accounts for Mexico’s transition from 1990 to 2015 with four estimated parameters and three time-varying wage components. Gender-neutral productivity explains roughly half of the fertility and family-structure decline; declining gender-biased productivity and human-capital convergence drive most of the gender income and welfare convergence. The model’s out-of-sample 2023 fertility prediction is 1.93, compared with the realized value of 1.91. A dynamic extension, in which boys from single-parent homes accumulate less human capital, shows how a technological shock can generate self-reinforcing, cross-generational declines in marriage and fertility.

The results point to two broader lessons. First, sustained progress along all three dimensions—fertility, family structure, and gender equality—is possible only when both spouses gain from the reform. Policies that quietly rely on one partner to absorb the cost of the other’s gains tend to fail at precisely the participation margin they need to move. Second, the welfare effects of family policy cannot be assessed in partial equilibrium. The within-family transfers that clear the marriage market respond endogenously to policy, and those responses can reverse the sign of the welfare effect that a partial-equilibrium analysis would predict.

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# Appendix

## A Additional Empirical Results

### A.1 OECD Time-Series Trends

Figure A.1 displays fertility, nonmarital births, and gender earnings gaps for OECD countries with at least 25 complete-case observations. This restriction selects longer series for readability.

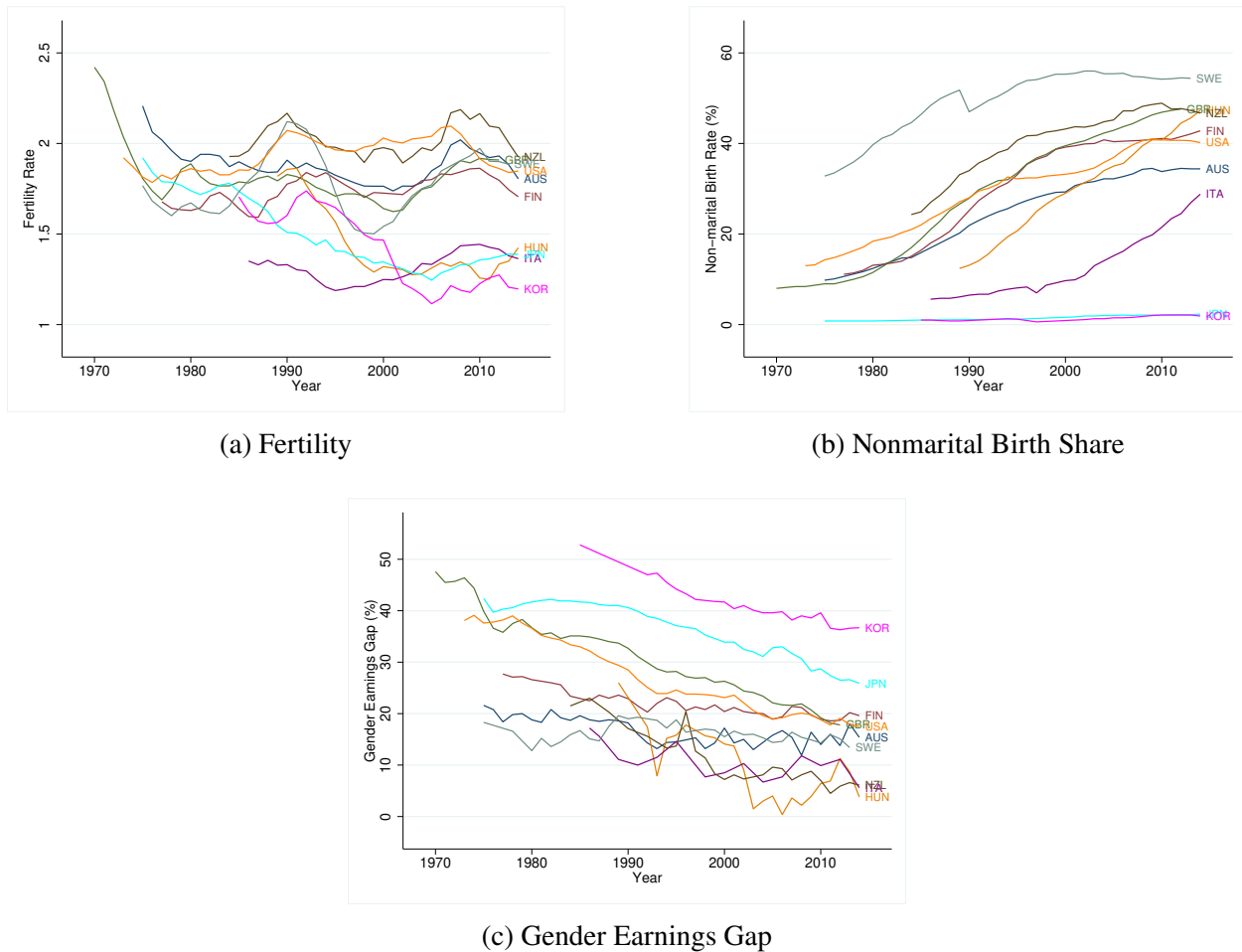
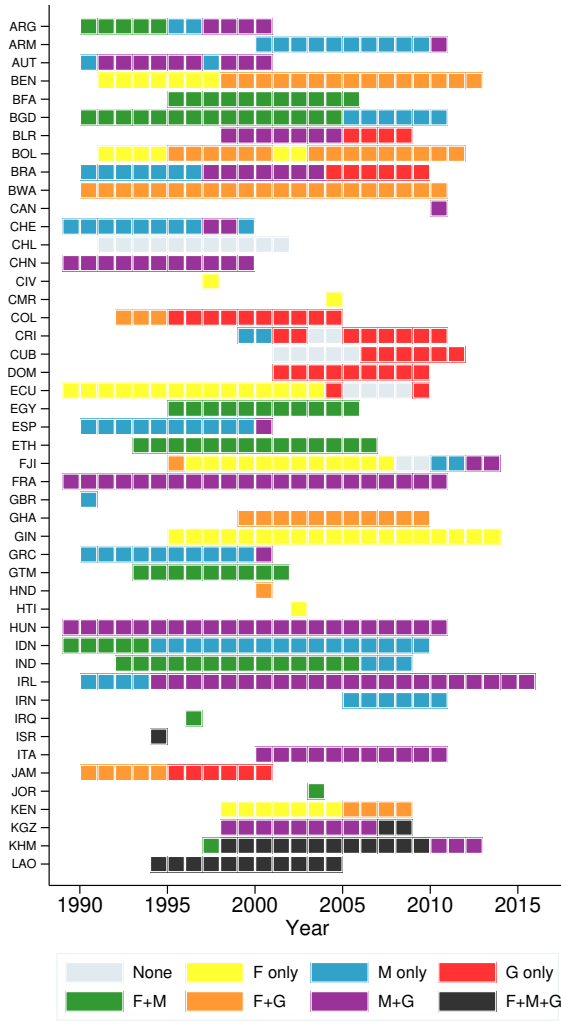


Figure A.1. Longer Country Series in the OECD Sample

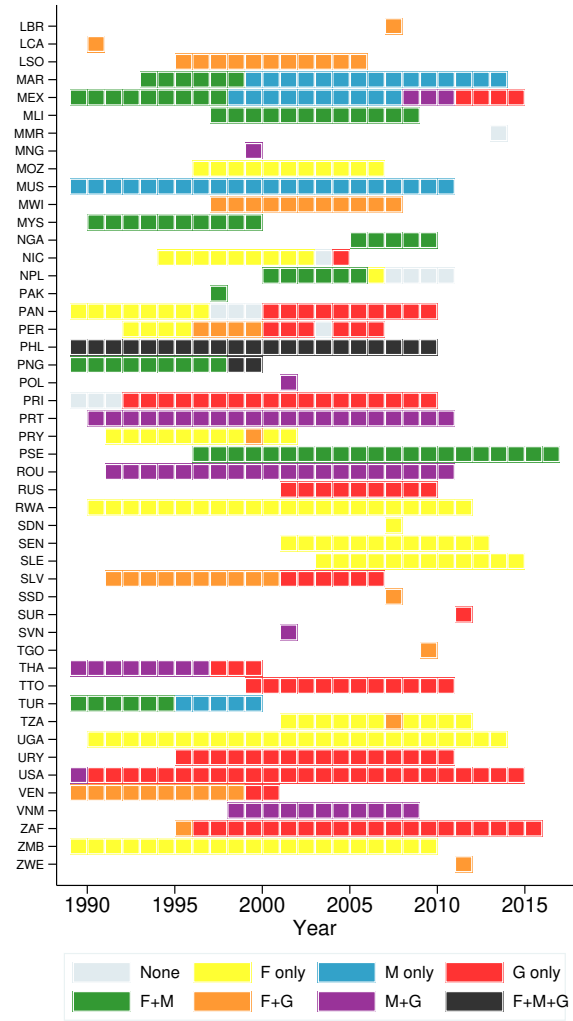
*Notes:* Countries have at least 25 observations in the complete-case OECD sample. The full panel spans 1970–2017, with country-specific coverage. Fertility is children per woman; the nonmarital birth share and earnings gap are in percentage points.

### A.2 World Sample Transition Paths

Figure A.2 traces transition paths for all countries in the World sample. Because of the large number of countries, the figure is split into two panels.



(a) World: panel 1



(b) World: panel 2

Figure A.2. Transition Paths by Country (World Sample)

Notes: Temporal transitions in the three categories (F = high fertility, M = dual parenthood, G = gender income equality) for all countries in the World sample, 1990 to 2015.

### A.3 U.S. Commuting Zone Analysis

The commuting-zone exercise uses 656 observations from year 2000. Fifty zones attain all three median-defined objectives, giving a joint share of 7.62 percent against an independence benchmark of 12.50 percent. The difference is  $-4.88$  percentage points, with a bootstrap percentile interval of  $[-6.78, -3.16]$  percentage points.

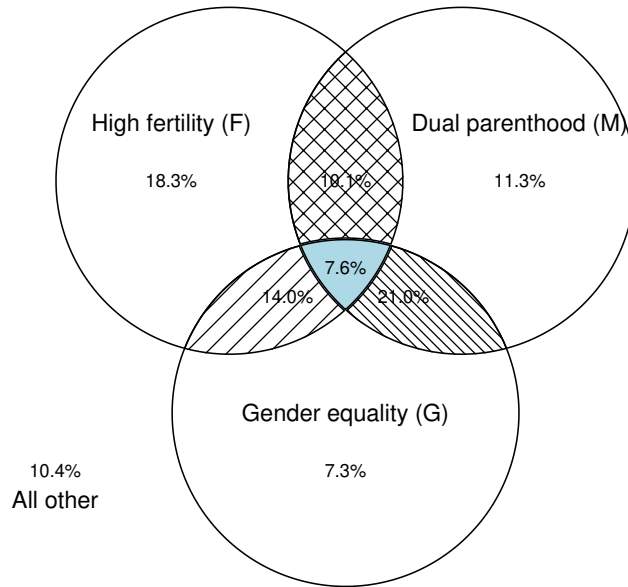


Figure A.3. Joint Attainment Across U.S. Commuting Zones: Legacy Measures

*Notes:* The sample contains 656 commuting zones in 2000. F uses the crude birth rate, M the reversed single-mother share, and G the reversed male-to-female earnings ratio. Thresholds are sample medians.

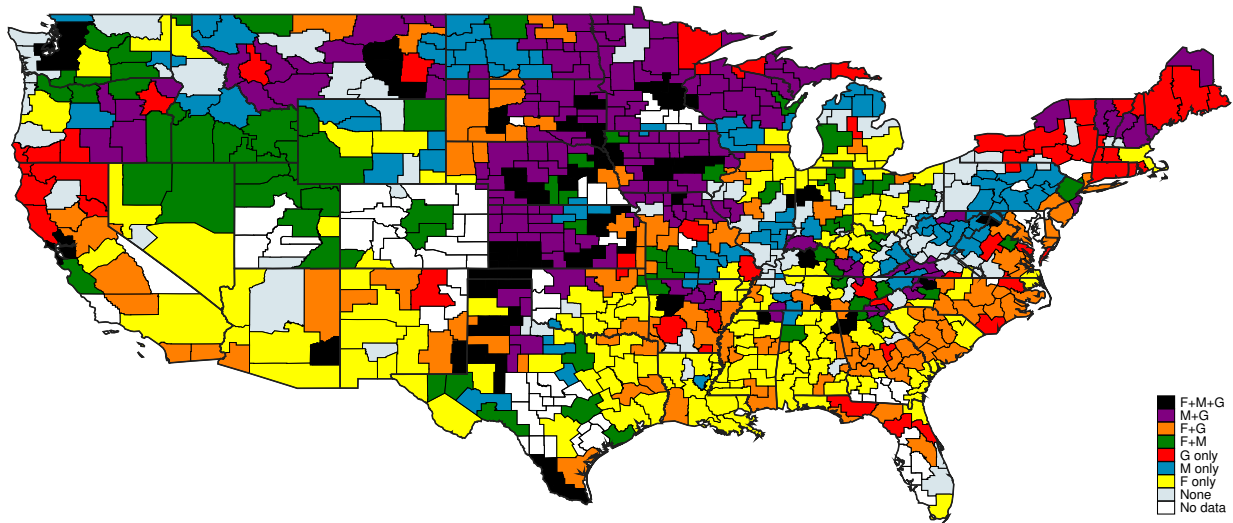


Figure A.4. Commuting-Zone Attainment Categories

*Notes:* Categories correspond to the median-defined indicators. The map applies the geographic display restrictions in the analysis code; numerical statistics use the 656 complete-case observations.

#### A.4 Subsamples by Income and Human Capital

Figure A.5 presents Venn diagrams for OECD subsamples stratified by income and human capital. Figure A.6 presents the corresponding World subsample figures. Outcome cutoffs are re-estimated within each subgroup. The subgroup Venn diagrams show the same central pattern: the trinity overlap is small relative to the product of the subgroup-specific marginal attainment rates. For example, in the OECD high-income subgroup, the observed trinity share is 5.28 percent against an independence benchmark of 12.29 percent; in the low-income subgroup, it is 3.05 percent against 12.53 percent. In the World high-income subgroup, the observed share is 0.92 percent against 12.48 percent, while in the low-income subgroup it is 6.45 percent against 12.43 percent.

#### A.5 Robustness to Thresholds, Measurement, and Sample Composition

##### Percentile thresholds

Figure A.7 and Table A.1 vary the common percentile cutoff from 25 to 75. At every cutoff, the independence benchmark is the product of the observed marginal attainment rates. This accommodates ties and avoids imposing the approximation  $(1 - q/100)^3$ .

| Sample      | Observed / empirical independence benchmark |          |          |          |          |          |          |
|-------------|---------------------------------------------|----------|----------|----------|----------|----------|----------|
|             | $q = 25$                                    | $q = 30$ | $q = 40$ | $q = 50$ | $q = 60$ | $q = 70$ | $q = 75$ |
| OECD        | 0.80                                        | 0.67     | 0.50     | 0.20     | 0.20     | 0.26     | 0.45     |
| World       | 0.82                                        | 0.70     | 0.48     | 0.35     | 0.39     | 0.36     | 0.06     |
| U.S. States | 0.87                                        | 0.81     | 0.81     | 0.50     | 0.76     | 1.13     | 1.64     |
| U.S. CZ     | 0.93                                        | 0.86     | 0.75     | 0.61     | 0.59     | 0.62     | 0.39     |

Table A.1. Sensitivity to Percentile Cutoffs

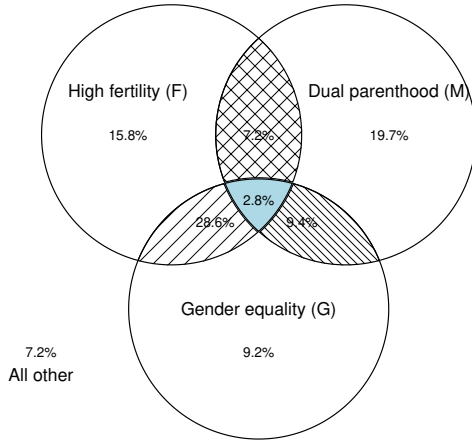
*Notes:* Ratios use unrounded observed and benchmark probabilities. The threshold grid is descriptive; the baseline bootstrap intervals do not provide simultaneous inference across cutoffs.

Ratios remain below one throughout the OECD, World, and legacy commuting-zone samples. In the state sample, the ratios exceed one at both the 70th and 75th percentiles. Each of those upper-tail overlaps contains one state observation, illustrating the sensitivity of sparse-cell comparisons in the small cross-section.

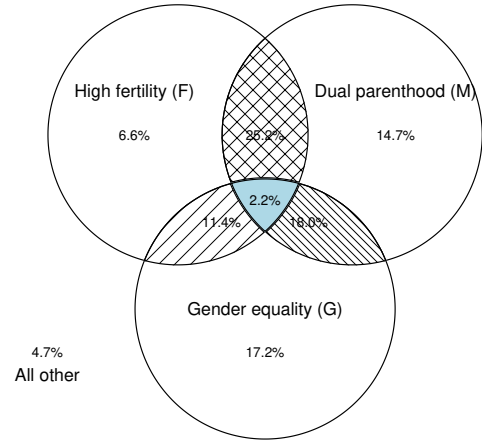
##### Continuous rank products

Let  $r_i^F$ ,  $r_i^M$ , and  $r_i^G$  be average ranks, retaining ties. Define

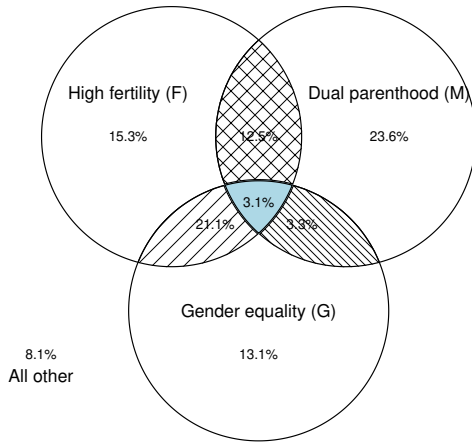
$$U_i^j = \frac{r_i^j - \frac{1}{2}}{N}, \quad j \in \{F, M, G\}.$$



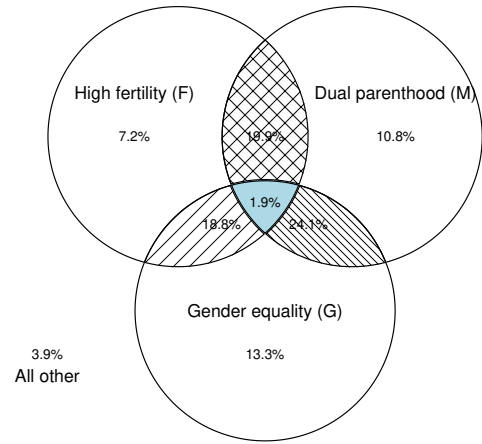
(a) High GDP per Capita



(b) Low GDP per Capita



(c) High Years of Schooling



(d) Low Years of Schooling

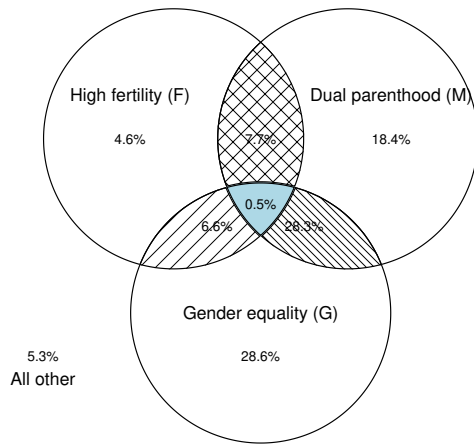
Figure A.5. Trade-off by Income and Human Capital Levels (OECD Sample)

Notes: Subsamples are split at median GDP per capita and median average years of schooling. Indicators are redefined at subsample medians in each case. Data cover 37 OECD countries from 1970 to 2017.

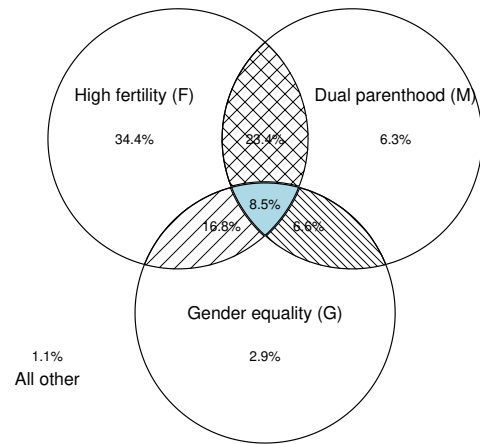
Each marginal mean is exactly one-half under this convention, including with ties. The rank-product statistic and its benchmark are

$$R = \frac{1}{N} \sum_i U_i^F U_i^M U_i^G, \quad R^{\text{ind}} = \bar{U}^F \bar{U}^M \bar{U}^G = \frac{1}{8}.$$

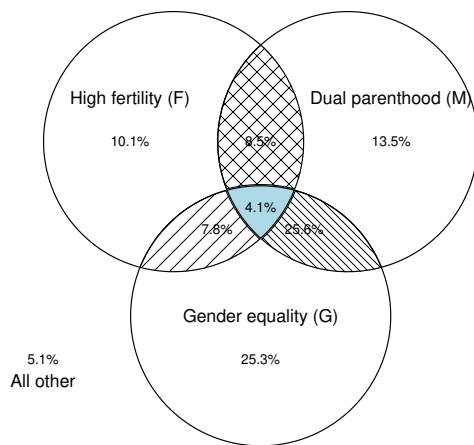
Table A.2 reports  $R$  and  $R - R^{\text{ind}}$ . The rank-product differences are negative in every sample.



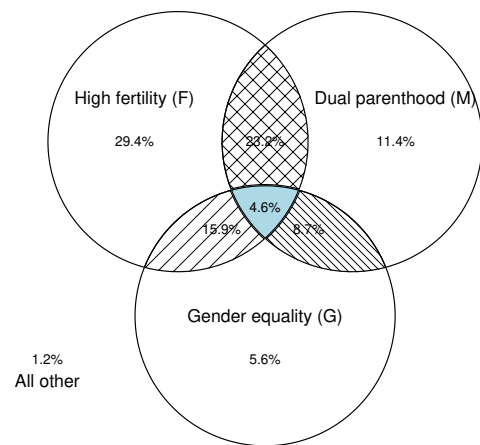
(a) High GDP per Capita



(b) Low GDP per Capita



(c) High Years of Schooling



(d) Low Years of Schooling

Figure A.6. Trade-off by Income and Human Capital Levels (World Sample)

Notes: Subsamples are split at median GDP per capita and median average years of schooling. Indicators are redefined at subsample medians in each case. Data cover 95 economies from 1990 to 2015.

### Absolute thresholds

Table A.3 uses absolute cutoffs. All reported joint-attainment ratios are below one.

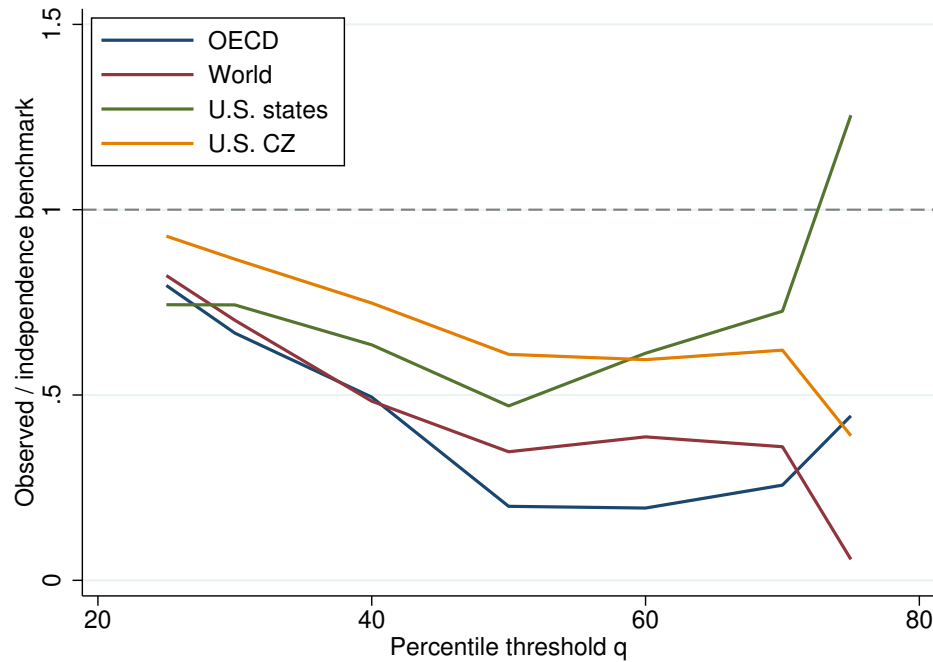


Figure A.7. Joint Attainment Across Percentile Thresholds

*Notes:* The vertical axis is observed joint attainment divided by the product of observed marginal attainment rates. Indicators equal one when the objective-aligned outcome strictly exceeds its sample percentile cutoff.

| Sample      | Rank product | Difference | Bootstrap SE | 95% interval       | p-value   |
|-------------|--------------|------------|--------------|--------------------|-----------|
| OECD        | 0.088        | -0.037     | 0.006        | $[-0.048, -0.023]$ | $< 0.001$ |
| World       | 0.084        | -0.041     | 0.005        | $[-0.051, -0.029]$ | $< 0.001$ |
| U.S. States | 0.104        | -0.021     | 0.011        | $[-0.040, 0.003]$  | 0.068     |
| U.S. CZ     | 0.106        | -0.019     | 0.003        | $[-0.024, -0.013]$ | $< 0.001$ |

Table A.2. Continuous Rank-Product Comparisons

*Notes:* Difference is the mean rank product minus 0.125. Ranks are recalculated within each bootstrap sample. Intervals and  $p$ -values come from the 999-replication re-estimated-cutoff bootstrap runs. The rank statistic itself is independent of binary cutoff choices.

## Alternative outcome measures

Alternative measures can change both the outcome definition and the available sample. Table A.4 separates those effects by reporting the main and alternative definitions on their common complete-case observations. Medians are recalculated for each definition on that common sample.

All common-sample ratios are below one. Some alternative definitions generate identical binary distributions on their common sample, as in the nonmarital-birth comparisons; those cases provide limited independent measurement variation.

| Sample      | Specification                                                         | <i>N</i> | Observed | Independent | Ratio |
|-------------|-----------------------------------------------------------------------|----------|----------|-------------|-------|
| OECD        | TFR > 2.1, nonmarital share < 30, gap < 15                            | 721      | 0.004    | 0.009       | 0.48  |
| OECD        | TFR > 1.8, nonmarital share < 40, gap < 20                            | 721      | 0.080    | 0.151       | 0.53  |
| World       | TFR > 2.1, both parents > .75, female share > .35                     | 1,130    | 0.025    | 0.099       | 0.25  |
| World       | TFR > 2.5, both parents > .80, female share > .40                     | 1,130    | 0.002    | 0.024       | 0.07  |
| U.S. States | Fertility > 56.9, single-mother share < .17, female/male ratio > .744 | 51       | 0.078    | 0.123       | 0.64  |
| U.S. CZ     | CBR > .012, single-mother share < .20, ratio < 1.80                   | 656      | 0.133    | 0.178       | 0.75  |
| U.S. CZ     | CBR > .013, single-mother share < .18, ratio < 1.60                   | 656      | 0.005    | 0.013       | 0.35  |

Table A.3. Joint Attainment Under Absolute Thresholds

*Notes:* Observed and independent columns are probabilities. The independence benchmark is the product of observed marginal attainment rates. OECD nonmarital shares and earnings gaps are in percentage points. CZ crude birth rates in this table are births per person, so .012 corresponds to 12 per 1,000 population.

| Sample | Replacement                         | <i>N</i> | Main definition |             |       | Alternative definition |       |
|--------|-------------------------------------|----------|-----------------|-------------|-------|------------------------|-------|
|        |                                     |          | Observed        | Independent | Ratio | Observed               | Ratio |
| OECD   | Alternative nonmarital-birth series | 89       | 0.079           | 0.126       | 0.62  | 0.079                  | 0.62  |
| OECD   | Female income share, new series     | 610      | 0.054           | 0.125       | 0.43  | 0.038                  | 0.30  |
| OECD   | Female income share, old series     | 610      | 0.054           | 0.125       | 0.43  | 0.043                  | 0.34  |
| World  | Reversed single-mother share        | 1,130    | 0.043           | 0.125       | 0.35  | 0.057                  | 0.45  |
| World  | Female income share, old series     | 1,130    | 0.043           | 0.125       | 0.35  | 0.047                  | 0.38  |
| World  | Alternative nonmarital-birth series | 118      | 0.017           | 0.125       | 0.14  | 0.017                  | 0.14  |

Table A.4. Alternative Measures on Common Samples

*Notes:* Observed and independent columns are probabilities. Every ratio uses the empirical independence benchmark for its own outcome definition.

## Excluding interpolation

Table A.5 retains observations identified as source-observed for the relevant interpolated inputs. This reduces the OECD sample from 721 to 551 observations and the World sample from 1,130 to 216. All 37 OECD countries and all 95 World economies remain represented.

Both attainment shortfalls retain confidence intervals below zero. Rank-product differences are also negative, with percentile intervals of  $[-0.047, -0.027]$  for OECD and  $[-0.049, -0.0267]$  for

| Sample                 | <i>N</i> | Observed (%) | Independent (%) | Difference (pp) | 95% interval (pp) | <i>p</i> -value |
|------------------------|----------|--------------|-----------------|-----------------|-------------------|-----------------|
| OECD, source-observed  | 551      | 3.63         | 12.34           | −8.71           | [−11.87, −4.97]   | < 0.001         |
| World, source-observed | 216      | 4.63         | 12.50           | −7.87           | [−11.46, −3.72]   | < 0.001         |

Table A.5. Joint Attainment After Excluding Script-Interpolated Observations

*Notes:* The OECD restriction checks fertility, nonmarital births, and earnings gaps. The World restriction checks fertility and children’s coresidence with both parents; the baseline female income share enters directly from the assembled dataset. These restrictions concern interpolation performed by this script and leave any upstream source imputation unassessed. Bootstrap inference uses 999 country-clustered replications with re-estimated cutoffs.

World.

### Leave-one-country-out sensitivity

The analysis omits each country in turn and re-estimates the cutoffs on the remaining sample. The joint-attainment difference remains negative in every iteration. In the OECD sample, it ranges from −10.54 to −8.84 percentage points; in the World sample, it ranges from −9.67 to −7.75 percentage points. Rank-product differences likewise remain negative throughout. These results show that the direction of the pooled shortfall survives removal of any individual country.

## A.6 Continuous Interaction Regressions

Continuous regressions offer a supplementary view of how the association between two outcomes varies with the third. I standardize the objective-aligned variables within each estimation sample and rotate the dependent variable:

$$z_F = \beta_0 + \beta_1 z_M + \beta_2 z_G + \beta_3 z_M z_G + \varepsilon,$$

$$z_M = \beta_0 + \beta_1 z_F + \beta_2 z_G + \beta_3 z_F z_G + \varepsilon,$$

$$z_G = \beta_0 + \beta_1 z_F + \beta_2 z_M + \beta_3 z_F z_M + \varepsilon.$$

Country-panel specifications include country and year fixed effects and country-clustered standard errors. The controlled specification adds log GDP per capita and the dataset’s human-capital index. A matched specification uses the same controls-complete sample without adding those controls. Quadratic specifications add squared terms for the two continuous regressors.

These interaction coefficients describe conditional associations under the specified functional form. They supplement the pooled-distribution comparisons and have a separate interpretation from interaction information or the binary pairwise-benchmark residual.

Table A.6 presents the regression results. Overall, the regressions show setting- and specification-dependent conditional associations.

| Sample      | Specification | Outcome | $N$   | Interaction | SE     | $p$ -value | Holm $p$ |
|-------------|---------------|---------|-------|-------------|--------|------------|----------|
| OECD        | Controlled FE | F       | 721   | −0.0523     | 0.0781 | 0.5070     | 0.5070   |
| OECD        | Controlled FE | M       | 721   | 0.1617      | 0.0405 | 0.0003     | 0.0009   |
| OECD        | Controlled FE | G       | 721   | −0.0976     | 0.0544 | 0.0810     | 0.1620   |
| World       | Controlled FE | F       | 1,033 | −0.0316     | 0.0575 | 0.5846     | 1.0000   |
| World       | Controlled FE | M       | 1,033 | 0.0451      | 0.0608 | 0.4605     | 1.0000   |
| World       | Controlled FE | G       | 1,033 | −0.0034     | 0.0550 | 0.9513     | 1.0000   |
| U.S. States | Baseline      | F       | 51    | 0.0225      | 0.0632 | 0.7229     | 1.0000   |
| U.S. States | Baseline      | M       | 51    | 0.0038      | 0.1613 | 0.9815     | 1.0000   |
| U.S. States | Baseline      | G       | 51    | 0.1734      | 0.1796 | 0.3393     | 1.0000   |
| U.S. CZ     | Baseline      | F       | 656   | −0.1632     | 0.0515 | 0.0016     | 0.0016   |
| U.S. CZ     | Baseline      | M       | 656   | −0.1493     | 0.0349 | < 0.001    | < 0.001  |
| U.S. CZ     | Baseline      | G       | 656   | −0.1542     | 0.0300 | < 0.001    | < 0.001  |

Table A.6. Continuous Interaction Regressions

*Notes:* Each interaction is the product of the two standardized outcomes other than the dependent variable. OECD controlled regressions contain 37 country clusters; World controlled regressions contain 86. U.S. cross-sections use heteroskedasticity-robust standard errors. Holm adjustment covers the three rotated outcomes within each sample and specification.

## B Additional Figures

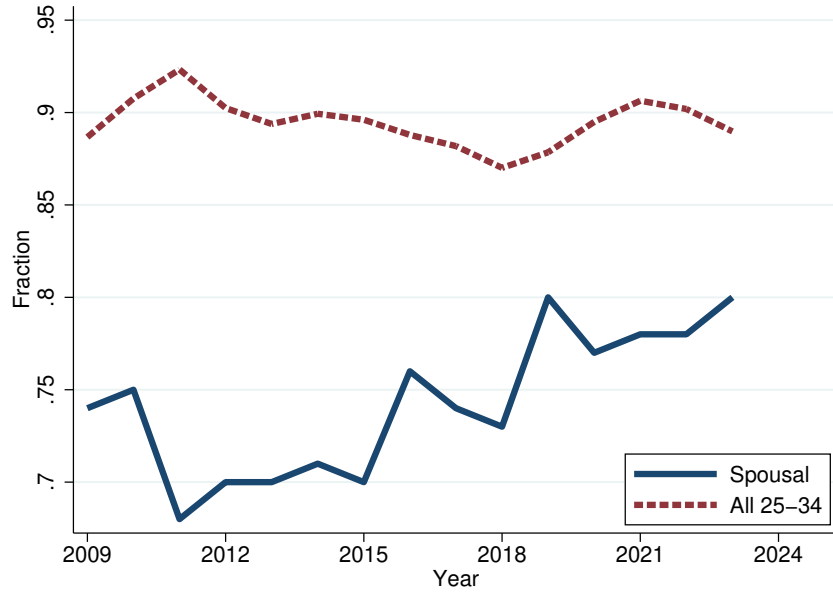
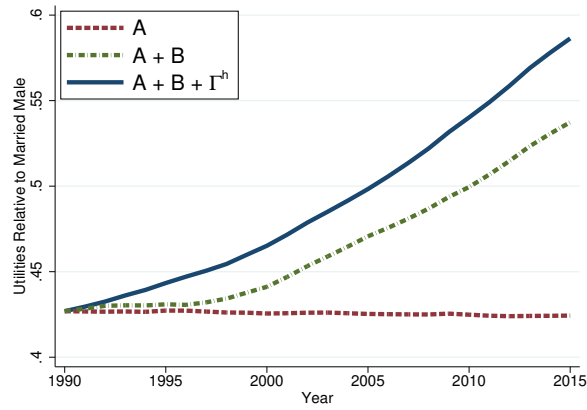
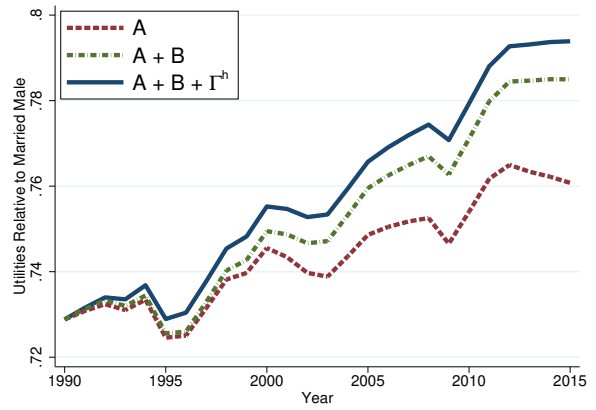


Figure A.8. Gender Earnings Gaps in the U.S.

*Notes:* The spousal earnings gap uses ACS women aged 18–39, not enrolled in school, childless, and in a first marriage within the past year to a husband also not enrolled in school. The overall gap for full-time wage and salary workers aged 25–34 is from the BLS Annual Highlights of Women’s Earnings.



(a) Relative Utility:  $V^Q,s / V^O,m$



(b) Relative Utility:  $V^O,s / V^O,m$

Figure A.9. Additional Results on Relative Utility

*Notes:* “ $A_t$  only” varies gender-neutral productivity; “ $A_t + B_t$ ” additionally varies residual gender bias; “All” is the full model. Drivers outside each experiment are fixed at their 1990 values. Utility ratios exclude taste shocks.

## C Proofs for the Baseline Model

### C.1 Marriage Probabilities and the Husband's Participation Relation

Let the independent shocks have distribution  $F(\varepsilon) = \exp(-\varepsilon^{-\theta})$  for  $\varepsilon > 0$ . An individual marries if  $\varepsilon^m V^{g,m} \geq \varepsilon^s V^{g,s}$ . Integrating over  $\varepsilon^s = x$  gives

$$\begin{aligned} \Pr(\varepsilon^m V^{g,m} < \varepsilon^s V^{g,s}) &= \int_0^\infty F\left(x \frac{V^{g,s}}{V^{g,m}}\right) dF(x) \\ &= \frac{(V^{g,s})^\theta}{(V^{g,m})^\theta + (V^{g,s})^\theta}. \end{aligned}$$

Taking the complement yields (2). Strict monotonicity in the utility ratio gives (3).

Single men's value is  $(1 - \beta)^{1/r} w^{\sigma^g}$ . Therefore

$$\frac{V^{\sigma^g,m}}{V^{\sigma^g,s}} = \lambda \left[ (1 - \alpha)^r + \frac{\beta}{1 - \beta} \left( \frac{n^m}{w^{\sigma^g}} \right)^r \right]^{1/r}. \quad (\text{C.1})$$

Setting this equal to  $(\mathcal{M}/(1 - \mathcal{M}))^{1/\theta}$  gives the conditional participation schedule

$$\alpha = 1 - \left[ \left( \frac{1}{\lambda} \left( \frac{\mathcal{M}}{1 - \mathcal{M}} \right)^{1/\theta} \right)^r - \frac{\beta}{1 - \beta} \left( \frac{n^m}{w^{\sigma^g}} \right)^r \right]^{1/r},$$

where its bracket is positive. At zero fertility this gives  $\alpha = 1 - \lambda^{-1}(\mathcal{M}/(1 - \mathcal{M}))^{1/\theta}$ , which equals zero only when  $\mathcal{M}/(1 - \mathcal{M}) = \lambda^\theta$ . This schedule describes the husband's willingness to participate conditional on fertility; the wife's household problem determines fertility.

### C.2 Household Choices and Proof of Lemma 2

*Proof.* For a woman with budget  $c + w^\varnothing \chi n = I$ , the interior first-order condition is

$$\frac{u_n}{u_c} = \frac{\beta}{1 - \beta} \left( \frac{c}{n} \right)^{1/\rho} = w^\varnothing \chi.$$

It follows that

$$\frac{c}{n} = \left( \frac{(1 - \beta)w^\varnothing \chi}{\beta} \right)^\rho,$$

and hence

$$n \left[ \left( \frac{(1 - \beta)w^\varnothing \chi}{\beta} \right)^\rho + w^\varnothing \chi \right] = I.$$

Setting  $I = w^{\varnothing}$  in singlehood and  $I = w^{\varnothing} + \alpha w^{\sigma}$  in marriage gives (6). Both consumption and fertility scale by  $1 + \alpha\Gamma^w$  across states. Homogeneity therefore implies

$$\frac{V_{\varnothing,m}^{\varnothing}}{V_{\varnothing,s}^{\varnothing}} = \lambda(1 + \alpha\Gamma^w).$$

At the equilibrium of Lemma 1,  $\alpha > 0$ , so marital fertility exceeds single fertility. Substitution into (2) proves the marriage-rate expression in (9).  $\square$

### C.3 Proof of Lemma 1

*Proof.* Combine (C.1), the wife's utility ratio, and (3). Canceling  $\lambda$  and raising both sides to power  $r$  gives

$$(1 - \alpha)^r + \frac{\beta}{1 - \beta} \left( \frac{n^m}{w^{\sigma}} \right)^r = (1 + \alpha\Gamma^w)^r.$$

Substitute  $n^m = (1 + \alpha\Gamma^w)/C$  to obtain

$$(1 - \alpha)^r = K(1 + \alpha\Gamma^w)^r.$$

Assumption 1 implies  $0 < K < 1$ . On  $-1/\Gamma^w < \alpha < 1$ , taking positive roots is valid:

$$1 - \alpha = K^{1/r}(1 + \alpha\Gamma^w).$$

This linear equation has the unique solution in (8), and substitution gives marital fertility.

The time restriction in (7) guarantees positive married female labor supply. Single female labor supply is positive because  $C > \chi$ , male consumption is positive because  $\alpha < 1$ , and female consumption is positive by the first-order condition. CES preferences are strictly quasiconcave on the positive domain, so the feasible interior solutions are unique household optima.

Every interior equilibrium satisfies the preceding equivalent equations. The constructed solution is therefore the unique interior equilibrium. The parameters  $\lambda$  and  $\theta$  affect participation, while the transfer and fertility formulas depend on the remaining primitives.  $\square$

### C.4 Aggregate Formulas and the Empirical Family-Structure Mapping

For compactness define

$$g \equiv K^{1/r}, \quad q \equiv 1 + \alpha^*\Gamma^w = \frac{1 + \Gamma^w}{1 + g\Gamma^w} > 1.$$

Then

$$n^s = \frac{1}{C}, \quad n^m = \frac{q}{C}, \quad \mathcal{M} = \frac{(\lambda q)^\theta}{1 + (\lambda q)^\theta}, \quad n = \frac{1 + \mathcal{M}(q - 1)}{C}. \quad (\text{C.2})$$

The child-weighted empirical counterpart of marriage is

$$\frac{\mathcal{M}n^m}{n} = \frac{\mathcal{M}q}{1 + \mathcal{M}(q - 1)}.$$

This expression uses the existing outcome notation. Its partial derivatives are

$$\frac{\partial}{\partial \mathcal{M}} \frac{\mathcal{M}q}{1 + \mathcal{M}(q - 1)} = \frac{q}{[1 + \mathcal{M}(q - 1)]^2} > 0, \quad \frac{\partial}{\partial q} \frac{\mathcal{M}q}{1 + \mathcal{M}(q - 1)} = \frac{\mathcal{M}(1 - \mathcal{M})}{[1 + \mathcal{M}(q - 1)]^2} > 0. \quad (\text{C.3})$$

Thus marriage and its child-based counterpart move together whenever  $\mathcal{M}$  and  $q$  move in the same direction, or one is unchanged and the other moves strictly. This applies to the individual-policy results. The relationship concerns their comparative statics; their levels reflect different population weights.

### C.5 Proof of Proposition 1

*Proof.* The objects  $C$ ,  $q$ , and household fertilities are independent of  $\lambda$ . Therefore

$$\mathcal{M}_\lambda = \frac{\theta \mathcal{M}(1 - \mathcal{M})}{\lambda} > 0, \quad n_\lambda = (n^m - n^s) \mathcal{M}_\lambda > 0.$$

With wages and  $\chi$  fixed,  $\Gamma^y = \Gamma^w / (1 - \chi n)$  rises. Since  $q$  is fixed, (C.3) also shows that the empirical family-structure counterpart rises.  $\square$

### C.6 Proof of Proposition 2

*Proof.* At fixed  $w^\sigma$ , an increase in  $w^\varnothing$  raises  $C$ ,  $K$ , and  $g$ , and lowers  $\Gamma^w$ . Since

$$q_{\Gamma^w} = \frac{1 - g}{(1 + g\Gamma^w)^2} > 0, \quad q_g = -\frac{(1 + \Gamma^w)\Gamma^w}{(1 + g\Gamma^w)^2} < 0,$$

both changes reduce  $q$ , so marriage falls. Both  $n^s = 1/C$  and  $n^m = q/C$  fall. Aggregate fertility satisfies

$$n_{w^\varnothing} = \mathcal{M}(n^m)_{w^\varnothing} + (1 - \mathcal{M})(n^s)_{w^\varnothing} + (n^m - n^s)\mathcal{M}_{w^\varnothing} < 0.$$

Female wages and female labor supply both rise, so  $\Gamma^y$  falls. The child-based family-structure measure falls because  $\mathcal{M}$  and  $q$  both fall. These conclusions hold for either sign of  $\alpha_{w^\varnothing}$ .  $\square$

### C.7 Proof of Proposition 3

*Proof.* Reducing  $\chi$  lowers  $C$ ,  $K$ , and  $g$  at fixed  $\Gamma^w$ , so  $q$  and marriage rise. Household fertilities rise, and the shift toward marriage reinforces aggregate fertility.

To determine the income response, write

$$\chi n = \frac{\chi}{C}[1 + \mathcal{M}(q - 1)], \quad \frac{\chi}{C} = \frac{1}{1 + \left(\frac{1-\beta}{\beta}\right)^\rho (w^\varnothing \chi)^{\rho-1}}.$$

Since  $\rho > 1$ , the first factor rises as  $\chi$  falls. The second also rises because  $\mathcal{M}$  and  $q$  increase and  $q > 1$ . Hence female labor supply falls and the income gap widens. The empirical family-structure counterpart rises by (C.3).  $\square$

### C.8 Conditional Feasibility and Coordinated Policy

*Derivation of (10).* Equation (9) gives

$$\frac{1}{\lambda} \left( \frac{\mathcal{M}}{1 - \mathcal{M}} \right)^{1/\theta} = 1 + \alpha^* \Gamma^w.$$

Since  $\alpha^* < 1$ ,

$$1 + \alpha^* \Gamma^w < 1 + \Gamma^w = 1 + \Gamma^y(1 - \chi n).$$

This proves the necessary restriction.  $\square$

**Proposition** (Local joint improvement using coordinated instruments). *At an interior baseline equilibrium, a sufficiently small increase in female wages combined with an appropriately chosen increase in the marriage benefit can raise fertility and marriage while narrowing the gender income gap.*

*Proof.* Hold  $w^{\sigma^\nearrow}$  and  $\chi$  fixed. Start with a female-wage increase and choose the marriage-benefit adjustment to keep aggregate fertility locally constant:

$$\frac{d\lambda}{dw^\varnothing} = -\frac{n_{w^\varnothing}}{n_\lambda} > 0.$$

The implicit function theorem applies because  $n_\lambda > 0$ . Household fertilities are independent of  $\lambda$ , so along this path

$$0 = \mathcal{M}(n^m)_{w^\varnothing} + (1 - \mathcal{M})(n^s)_{w^\varnothing} + (n^m - n^s) \frac{d\mathcal{M}}{dw^\varnothing}.$$

The first two terms are negative and  $n^m > n^s$ , implying  $d\mathcal{M}/dw^\varnothing > 0$ . The income gap falls because the female wage rises while female labor supply remains constant.

Choose a slightly larger positive slope for  $\lambda$ . Aggregate fertility then rises strictly. Continuity preserves the strict marriage increase and income-gap decline for a sufficiently small change. Thus all three objectives improve.

The child-based measure improves as well. On the constant-fertility path it equals  $1 - (1 - \mathcal{M})n^s/n$ , which rises because marriage rises and single fertility falls. This strict increase persists under the sufficiently small additional marriage-benefit adjustment.  $\square$

## D Extended Model with Gender-Differentiated Childcare

### D.1 Household Solutions and Proof of Lemma 3

Define the monetary fertility price and preference ratio

$$P \equiv \tilde{w}^\varnothing \tilde{\chi}^\varnothing = w^\varnothing \chi^\varnothing + \alpha w^\sigma \chi^\sigma, \quad k \equiv \frac{\beta}{1 - \beta}.$$

Throughout the derivative calculations, household allocations are interior,  $P > 0$ , and both spouses' consumption and labor supply are positive.

*Proof.* The wife's budget is  $c^{\varnothing,m} + Pn^m = \tilde{w}^\varnothing$ . The first-order condition yields

$$\frac{c^{\varnothing,m}}{n^m} = \left( \frac{P}{k} \right)^\rho.$$

Substitution gives

$$n^m = \frac{\tilde{w}^\varnothing}{P + (P/k)^\rho}, \quad c^{\varnothing,m} = \frac{\tilde{w}^\varnothing (P/k)^\rho}{P + (P/k)^\rho}. \quad (\text{D.1})$$

Dividing the fertility denominator by  $\tilde{w}^\varnothing$  proves (13). Substitution into CES utility gives

$$V^{\varnothing,m} = \lambda(1 - \beta)^{1/r} \tilde{w}^\varnothing (1 + k^\rho P^{1-\rho})^{1/(\rho-1)},$$

which proves (14). The singlehood formula follows by the indicated substitutions.

At fixed effective resources, fertility and utility decrease with  $P$ . Consumption can be written as

$$c^{\varnothing,m} = \frac{\tilde{w}^\varnothing}{1 + k^\rho P^{1-\rho}},$$

which increases with  $P$  when  $\rho > 1$ . Lower fertility prices therefore raise fertility and utility while lowering consumption.  $\square$

An equivalent fertility-proportional representation is

$$V^{\varnothing,m} = \lambda n^m \Psi^m, \quad \Psi^m = \left[ (1 - \beta) \left( \frac{(1 - \beta)P}{\beta} \right)^{\rho-1} + \beta \right]^{1/r}.$$

The singlehood representation replaces  $P$  with  $w^{\varnothing}\chi$  and omits  $\lambda$ .

## D.2 Equilibrium Derivatives

Partial derivatives with respect to  $\delta$  hold  $\alpha$  fixed; primes denote total equilibrium derivatives. All expressions are evaluated at the initial equilibrium. Define

$$d \equiv w^{\varnothing} - \alpha w^{\sigma}, \quad v \equiv 1 - \chi^{\sigma} n^m > 0,$$

$$z \equiv \frac{P n^m}{\tilde{w}^{\varnothing}} \in (0, 1), \quad \eta \equiv \rho - (\rho - 1)z, \quad e \equiv \eta - z = \rho(1 - z) > 0.$$

Let

$$T \equiv \frac{u_n(c^{\sigma,m}, n^m)}{u_c(c^{\sigma,m}, n^m)} = k \left( \frac{c^{\sigma,m}}{n^m} \right)^{1/\rho},$$

$$R \equiv T - (1 - \alpha)w^{\sigma}\chi^{\sigma}, \quad Q \equiv c^{\sigma,m} + T n^m > 0, \quad \mathcal{B} \equiv 1 - \frac{\eta \chi^{\sigma} \tilde{w}^{\varnothing}}{P}.$$

Finally define

$$Z \equiv (w^{\sigma} + w^{\varnothing})v + \frac{R \chi^{\sigma} n^m \tilde{w}^{\varnothing} e}{P}, \quad J \equiv \frac{R d e}{P} - v(w^{\sigma} - w^{\varnothing}). \quad (\text{D.2})$$

Write the spouses' log marriage premiums as

$$f \equiv \log \frac{V^{\varnothing,m}}{V^{\varnothing,s}}, \quad h \equiv \log \frac{V^{\sigma,m}}{V^{\sigma,s}},$$

and let  $S \equiv f_{\alpha} - h_{\alpha}$ . Market clearing requires  $f = h$ .

**Lemma** (Local equilibrium responses). *At an interior equilibrium with  $Z \neq 0$ , the implicit function*

theorem gives a differentiable local equilibrium branch. Its responses satisfy

$$G \equiv f' = h' = \frac{n^m J}{Z}, \quad (\text{D.3})$$

$$\mathcal{M}' = \theta \mathcal{M}(1 - \mathcal{M})G, \quad (\text{D.4})$$

$$\alpha' = \frac{\tilde{w}^\varphi G - dn^m}{w^\sigma v}, \quad (\text{D.5})$$

$$\frac{(n^m)'}{n^m} = \frac{\mathcal{B}G + de/P}{v}. \quad (\text{D.6})$$

Equivalently,

$$\frac{(n^m)'}{n^m} = \frac{(de/P)(\tilde{w}^\varphi + Q) - \mathcal{B}n^m(w^\sigma - w^\varphi)}{Z}. \quad (\text{D.7})$$

*Proof.* The fertility formula gives

$$n_\alpha^m = \frac{n^m w^\sigma \mathcal{B}}{\tilde{w}^\varphi}, \quad n_\delta^m = \frac{\eta n^m d}{P}. \quad (\text{D.8})$$

The wife's envelope condition yields

$$f_\alpha = \frac{w^\sigma v}{\tilde{w}^\varphi} > 0, \quad f_\delta = \frac{dn^m}{\tilde{w}^\varphi}. \quad (\text{D.9})$$

A transfer change increases her net resources by  $w^\sigma v$ . Homogeneity and optimality imply  $u_c/u = 1/\tilde{w}^\varphi$ .

For the husband, Euler's theorem gives  $d \log u = (dc^{\sigma,m} + T dn^m)/Q$ . Therefore

$$h_\alpha = \frac{-w^\sigma v + Rn_\alpha^m}{Q}, \quad h_\delta = \frac{-(1 - \alpha)w^\sigma n^m + Rn_\delta^m}{Q}. \quad (\text{D.10})$$

Differentiating  $f = h$  gives

$$\alpha' = \frac{h_\delta - f_\delta}{S}, \quad G = \frac{f_\alpha h_\delta - h_\alpha f_\delta}{S}.$$

Using  $Q = (1 - \alpha)w^\sigma + Rn^m$ , substitution yields

$$S = \frac{w^\sigma Z}{\tilde{w}^\varphi Q}, \quad f_\alpha h_\delta - h_\alpha f_\delta = \frac{w^\sigma n^m J}{\tilde{w}^\varphi Q}.$$

Thus  $Z \neq 0$  implies  $S \neq 0$ , proving local regularity and (D.3). The participation rule implies (D.4). Solving  $G = f_\delta + f_\alpha \alpha'$  gives (D.5).

Finally,  $(n^m)' = n_\delta^m + n_\alpha^m \alpha'$  and the identity  $\eta v - z\mathcal{B} = e$  imply (D.6). Substituting  $G = n^m J/Z$  and using

$$Z + \mathcal{B}n^m R = v(\tilde{w}^\varnothing + Q)$$

gives (D.7). □

Since single fertility is fixed, aggregate responses are

$$n' = \mathcal{M}(n^m)' + \mathcal{M}'(n^m - n^s), \quad (\text{D.11})$$

$$(\mathcal{M}n^m)' = \mathcal{M}(n^m)' + \mathcal{M}'n^m. \quad (\text{D.12})$$

The child-based family-structure response is

$$\left(\frac{\mathcal{M}n^m}{n}\right)' = \frac{n^s [n^m \mathcal{M}' + \mathcal{M}(1 - \mathcal{M})(n^m)']}{n^2}.$$

Consequently, rising marriage and rising marital fertility imply an increase in this empirical counterpart. The mapping expresses the response using the existing theoretical outcomes.

### D.3 The Male-Side Compensated Interpretation

Consider an auxiliary transfer adjustment  $\alpha'_c$  that holds the wife's log marriage premium fixed:

$$0 = f_\delta + f_\alpha \alpha'_c, \quad \alpha'_c = -\frac{dn^m}{w^\sigma v}.$$

This auxiliary path isolates the husband's gain after compensating the wife; the actual equilibrium transfer adjusts to clear the marriage market. Fertility along the auxiliary path satisfies

$$\frac{(n^m)'_c}{n^m} = \frac{\eta d}{P} - \frac{n^m \mathcal{B} d}{\tilde{w}^\varnothing v} = \frac{de}{Pv}.$$

The husband's compensated log-premium derivative is

$$h'_c = h_\delta + h_\alpha \alpha'_c = \frac{n^m J}{vQ}. \quad (\text{D.13})$$

Hence Condition (18), which is  $J > 0$ , is exactly a positive husband-side gain after the transfer compensates the wife and fertility responds.

To relate this comparison to equilibrium, observe

$$G = \frac{f_\alpha}{S} h'_c.$$

Under Conditions (15) and (18), the proof below establishes  $S > 0$ , so a positive compensated male gain produces a positive common equilibrium premium. This gives the condition its bilateral interpretation: the reform creates room to make both sides more willing to marry.

#### D.4 Proof of Proposition 4

*Proof.* Condition (15) is  $d > 0$ , and Condition (18) is  $J > 0$ . Since  $w^{\sigma} > w^{\varphi}$  and  $v, e, P > 0$ , they imply  $R > 0$ . Consequently

$$Z = (w^{\sigma} + w^{\varphi})v + \frac{R\chi^{\sigma}n^m\tilde{w}^{\varphi}e}{P} > 0.$$

Thus  $S > 0$ , and the equilibrium is locally regular. Equations (D.3) and (D.4) imply  $G > 0$  and  $\mathcal{M}' > 0$ . Singlehood values are fixed, so both spouses' marital values and marriage-to-singlehood ratios rise.

For marital fertility, if  $\mathcal{B} \geq 0$ , (D.6) is positive because  $G > 0$  and  $de/P > 0$ . If  $\mathcal{B} < 0$ , both terms in the numerator of (D.7) are positive. Therefore  $(n^m)' > 0$  in either case.

The fixed-total-care assumption gives

$$P = w^{\varphi}\chi - d\chi^{\sigma} \leq w^{\varphi}\chi, \quad \tilde{w}^{\varphi} > w^{\varphi}.$$

The fertility formula is increasing in resources and decreasing in price, so  $n^m > n^s$ . Equation (D.11) implies  $n' > 0$ .

Define the increase in fathers' aggregate care as

$$X \equiv \mathcal{M}n^m + \chi^{\sigma}(\mathcal{M}n^m)'.$$

Because marriage and marital fertility rise,  $X > 0$ . Differentiating labor supplies gives

$$(l^{\sigma})' = -X < 0, \quad (l^{\varphi})' = X - \chi n'.$$

At fixed wages,

$$\frac{d \log \Gamma^y}{d\delta} = \frac{(l^{\sigma})'}{l^{\sigma}} - \frac{(l^{\varphi})'}{l^{\varphi}} = \frac{\chi n'}{l^{\varphi}} - X \left( \frac{1}{l^{\sigma}} + \frac{1}{l^{\varphi}} \right).$$

Condition (17) is exactly the requirement that this expression is negative. Thus the income gap narrows without requiring  $(l^{\varphi})' > 0$ . The three desired signs follow. Positive derivatives and strict interiority imply the same directional changes for sufficiently small feasible reforms. The child-based counterpart also rises by the formula following (D.12).  $\square$

## D.5 Redistribution Starting from the Baseline

At  $\chi^{\sigma} = 0$  and  $\chi^{\varphi} = \chi$ ,

$$P = w^{\varphi}\chi, \quad v = 1, \quad R = T, \quad \mathcal{B} = 1, \quad Z = w^{\sigma} + w^{\varphi}.$$

The response formulas simplify to

$$G = \frac{n^m}{w^{\sigma} + w^{\varphi}} \left[ \frac{T(w^{\varphi} - \alpha w^{\sigma})\rho(1-z)}{w^{\varphi}\chi} - (w^{\sigma} - w^{\varphi}) \right], \quad (\text{D.14})$$

$$\frac{(n^m)'}{n^m} = G + \frac{(w^{\varphi} - \alpha w^{\sigma})\rho(1-z)}{w^{\varphi}\chi}. \quad (\text{D.15})$$

Here

$$z = \frac{w^{\varphi}\chi n^m}{w^{\varphi} + \alpha w^{\sigma}}, \quad T = \frac{\beta}{1-\beta} \left( \frac{(1-\alpha)w^{\sigma}}{n^m} \right)^{1/\rho}.$$

The male-side condition is

$$T(w^{\varphi} - \alpha w^{\sigma})\rho(1-z) > w^{\varphi}\chi(w^{\sigma} - w^{\varphi}).$$

The income gap narrows under

$$\chi n' < \mathcal{M}n^m(1 + l^{\varphi}).$$

Marriage and aggregate fertility derivatives follow from (D.4) and (D.11).

## D.6 Proof of Proposition 5

*Proof.* The maintained regularity condition is  $S > 0$ , equivalent to  $Z > 0$ .

**Case (A).** If  $d \leq 0$ , then  $n_{\delta}^m \leq 0$  and  $f_{\delta} \leq 0$ . At the baseline,  $R = T > 0$ ,  $v = 1$ , and  $\mathcal{B} = 1$ , so

$$J = \frac{Tde}{P} - (w^{\sigma} - w^{\varphi}) < 0.$$

Thus  $G < 0$ ,  $\mathcal{M}' < 0$ , and, by (D.15),  $(n^m)' < 0$ , including when  $d = 0$ . Baseline marital fertility exceeds single fertility, so  $n' < 0$ . Equation (??) becomes

$$\frac{d \log \Gamma^y}{d\delta} = \frac{\chi n'}{l^{\varphi}} - \mathcal{M}n^m \left( 1 + \frac{1}{l^{\varphi}} \right) < 0.$$

The child-based counterpart falls because marriage and marital fertility both fall. With positive paternal care initially, the fixed-transfer signs remain valid, while determining the full equilibrium signs requires additional restrictions.

**Case (B).** Here  $f_\delta > 0$  and  $h_\delta \leq 0$ . Therefore

$$\alpha' = \frac{h_\delta - f_\delta}{S} < 0.$$

The marriage response is governed by  $G = n^m J/Z$ , and the fertility response by (D.6) or (D.7). These expressions incorporate the transfer adjustment together with the direct fertility effect.

**Case (C).** The first two conditions give rising marriage and fertility and  $X > 0$ . Failure of Condition (17) means

$$\chi n' \geq X \left( 1 + \frac{l^\varnothing}{l^\sigma} \right).$$

Using

$$\frac{d \log \Gamma^y}{d\delta} = \frac{\chi n'}{l^\varnothing} - X \left( \frac{1}{l^\sigma} + \frac{1}{l^\varnothing} \right),$$

this gives  $d \log \Gamma^y / d\delta \geq 0$ . □

## D.7 Different Total Childcare Requirements by Marital Status

The household and equilibrium derivative formulas remain valid if  $\chi^\varnothing + \chi^\sigma \neq \chi$ , provided both total requirements remain fixed during redistribution and household choices remain interior. The aggregate fertility conclusion additionally requires  $n^m \geq n^s$ ; with increasing marriage and strictly increasing marital fertility, this is sufficient for  $n' > 0$ .

Labor supplies are

$$l^\sigma = 1 - \chi^\sigma \mathcal{M} n^m, \quad l^\varnothing = 1 - \chi^\varnothing \mathcal{M} n^m - \chi(1 - \mathcal{M}) n^s.$$

Their derivatives are

$$\begin{aligned} (l^\sigma)' &= -\mathcal{M} n^m - \chi^\sigma (\mathcal{M} n^m)', \\ (l^\varnothing)' &= \mathcal{M} n^m - \chi^\varnothing (\mathcal{M} n^m)' + \chi n^s \mathcal{M}'. \end{aligned}$$

The weaker exact income-gap condition for this case is

$$\frac{\mathcal{M} n^m - \chi^\varnothing (\mathcal{M} n^m)' + \chi n^s \mathcal{M}'}{l^\varnothing} + \frac{\mathcal{M} n^m + \chi^\sigma (\mathcal{M} n^m)'}{l^\sigma} > 0.$$

The child-based counterpart still rises whenever marriage and marital fertility rise, since single fertility remains fixed.